

Unveiling Survival Dynamics in Heart Failure Patients: A Comprehensive Time-to-Event Analysis



A project submitted
for partial fulfillment of the requirements for the award of the
degree of

Master of Science

by

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Declaration

I, **Sayan Banerjee**, hereby declare that the project titled *Unveiling Survival Dynamics in Heart Failure Patients: A Comprehensive Time-to-Event Analysis* and the work presented therein are my own and original, carried out under the supervision of **Dr. Ashish Kumar Yadav, Centre for Interdisciplinary Mathematical Sciences**, Banaras Hindu University, during the period from March 2026 to May 2026.

This project has been submitted to the DST-Centre for Interdisciplinary Mathematical Sciences, Banaras Hindu University, in partial fulfilment of the requirements for the M.Sc. in Statistics & Computing course.

To the best of my knowledge and belief, the information and data presented in this project are authentic. This work has not been submitted to any other university or institution for the award of any degree, diploma, or fellowship, nor has it been published previously.

Place: Varanasi
Date: May, 2026

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CERTIFICATE

This is to certify that **Sayan Banerjee**, a student of the **Master of Science in Statistics & Computing**, Semester IV of the **DST-Centre for Interdisciplinary Mathematical Sciences**, Institute of Science, Banaras Hindu University, bearing Exam Roll No. **24419STC038** and Enrollment No. **482751** of 2024–2026, has successfully completed his project entitled “**Unveiling Survival Dynamics in Heart Failure Patients: A Comprehensive Time-to-Event Analysis**”.

The project was completed under my direct supervision and guidance for the partial fulfillment of the course curriculum of Master of Science in Statistics & Computing. He carried out the whole work with utmost sincerity, and the present report is the outcome of his sincere efforts and endeavors.

This work, either in the present form or in part in any other form, is not being submitted anywhere to the best of my knowledge.

Supervisor

Course Coordinator

Coordinator, DST-CIMS

Acknowledgements

I would like to express my sincere gratitude to my supervisor, **Dr. Ashish Kumar Yadav**, Professor, Department of Biostatistics, Institute of Medical Sciences (IMS), Banaras Hindu University, for his invaluable guidance, continuous support, and encouragement throughout the course of this project. His insights and expertise in the field of biostatistics greatly contributed to the successful completion of this work.

I am also thankful to the faculty members of the Department of Statistics for providing a strong academic foundation and necessary resources that helped me carry out this study effectively.

I extend my heartfelt thanks to my friends and classmates for their support, constructive discussions, and motivation during the course of this project.

Finally, I would like to express my deepest gratitude to my family for their constant encouragement, patience, and unwavering support, without which this work would not have been possible.

Abstract

Survival analysis plays a crucial role in medical research for understanding time-to-event outcomes and identifying key risk factors influencing patient prognosis. This study focuses on the analysis of survival patterns among patients with heart failure using the Heart Failure Clinical Records Dataset.

An extensive exploratory data analysis (EDA) was conducted to understand the structure, distribution, and relationships within the data. Kaplan–Meier survival curves were used to estimate survival probabilities, and log-rank tests were performed to compare survival distributions across different patient groups. The proportional hazards assumption was assessed using log-minus-log plots and Schoenfeld residuals.

A Cox Proportional Hazards (PH) model was initially fitted; however, violations of the proportional hazards assumption were observed for certain variables. To address this, an extended Cox model incorporating time-dependent covariates was implemented. Additionally, parametric Accelerated Failure Time (AFT) models, including Weibull, Log-Normal, and Log-Logistic models, were fitted and compared.

Model performance was evaluated using log-likelihood, Akaike Information Criterion (AIC), and concordance index (C-index). The extended Cox model demonstrated superior performance compared to both the standard Cox model and AFT models, indicating its effectiveness in capturing time-varying effects in the data.

The findings highlight the importance of advanced survival modeling techniques in accurately assessing patient risk and provide valuable insights into factors affecting survival in heart failure patients.

Keywords: Survival Analysis, Kaplan–Meier Estimator, Cox Proportional Hazards Model, Time-Dependent Covariates, AFT Models, Heart Failure

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CHAPTER 1

Introduction

Heart failure is a critical public health concern worldwide, associated with high morbidity and mortality rates. Accurate prediction of patient survival and identification of key risk factors are essential for improving clinical decision-making and optimizing treatment strategies.

Survival analysis provides a robust statistical framework for analyzing time-to-event data, particularly in the presence of censoring. Unlike traditional regression methods, survival models explicitly account for the time until an event occurs, making them highly suitable for medical prognosis studies.

This study focuses on the analysis of the *Heart Failure Clinical Records Dataset*, with the aim of understanding survival patterns and identifying significant clinical predictors influencing patient mortality. The Cox Proportional Hazards (Cox PH) model is employed as the primary modeling framework, along with its extensions to address violations of model assumptions.

In addition to model fitting, extensive diagnostic analysis is performed using Martingale residual plots and Cox–Snell residual plots to assess model adequacy. Further, the study explores partial effects of covariates, estimates survival probabilities, and predicts expected survival times to provide meaningful clinical insights.

1.1 Objectives

The main objectives of this study are as follows:

- To perform exploratory data analysis (EDA) to understand the structure and characteristics of the dataset.
- To estimate survival functions using non-parametric methods such as Kaplan–Meier estimation.
- To compare survival distributions across different groups using log-rank tests.
- To identify significant risk factors affecting patient survival using the Cox Proportional Hazards model.

- To assess the validity of the proportional hazards assumption and apply extended Cox models where necessary.
- To perform model diagnostics using Martingale residuals and Cox–Snell residuals.
- To analyze and interpret partial effects of covariates on survival.
- To estimate survival probabilities for patients based on fitted models.
- To predict expected survival times (lifetime) for individuals.
- To compare model performance and evaluate goodness-of-fit.

1.2 Scope of the Study

This study focuses on survival analysis of heart failure patients using clinical variables available in the dataset. The analysis is restricted to semi-parametric and extended Cox proportional hazards models.

The study does not incorporate external clinical data, advanced machine learning survival models, or real-time prediction systems. The findings are therefore limited to the given dataset and may not generalize to broader populations without further validation.

Despite these limitations, the study provides valuable insights into key risk factors, survival probabilities, and expected lifetimes of patients.

1.3 Organization of the Report

The remainder of this report is organized as follows:

- Chapter 2 describes the dataset and variables used in the study.
- Chapter 3 presents exploratory data analysis and initial findings.
- Chapter 4 - 6 discusses the methodology, including survival models used.
- Chapter 7 presents model results, diagnostics, and concludes the study with key findings and future scope.

CHAPTER 2

Dataset Description

The dataset used in this study is the **Heart Failure Clinical Records Dataset**, obtained from the UCI Machine Learning Repository. It contains clinical data of **299 patients** with heart failure, observed over a follow-up period.

Dataset Summary:

- Number of observations: 299
- Number of variables: 13
- Outcome: Survival time and death indicator
- No missing values reported

Variables

Variable	Type	Description
Age	Integer	Age (years)
Anaemia	Binary	Reduced red blood cells/hemoglobin
CPK	Integer	Creatinine phosphokinase level (mcg/L)
Diabetes	Binary	Diabetes status (0/1)
Ejection Fraction	Integer	% blood leaving heart per contraction
High BP	Binary	Hypertension indicator (0/1)
Platelets	Continuous	Platelet count (kiloplatelets/mL)
Serum Creatinine	Continuous	Kidney function (mg/dL)
Serum Sodium	Integer	Sodium level (mEq/L)
Sex	Binary	Gender (0=female, 1=male)
Smoking	Binary	Smoking status (0/1)
Time	Integer	Follow-up duration (days)
DEATH_EVENT	Binary	Event indicator (0=alive, 1=death)

CHAPTER 3

Exploratory Data Analysis

Before performing survival analysis, an extensive exploratory data analysis (EDA) was carried out to understand the structure, quality, and characteristics of the dataset. The aim of this step was to summarize the data descriptively, identify possible outliers, inspect the distribution of variables, and explore the relationship between predictors and the survival outcome.

3.1 Dataset Overview

The dataset used in this study is the *Heart Failure Clinical Records Dataset*. It contains clinical information of patients with heart failure and is suitable for both classification and survival analysis. The dataset consists of 299 observations and 13 variables. Among these, `time` represents the follow-up duration, and `DEATH_EVENT` is the event indicator used for survival analysis.

3.2 Study Variables

The variables available in the dataset are listed below:

- `age`: Age of the patient
- `anaemia`: Whether the patient has anaemia (0 = No, 1 = Yes)
- `creatinine_phosphokinase`: Level of creatinine phosphokinase enzyme in blood
- `diabetes`: Whether the patient has diabetes (0 = No, 1 = Yes)
- `ejection_fraction`: Percentage of blood leaving the heart at each contraction
- `high_blood_pressure`: Whether the patient has high blood pressure (0 = No, 1 = Yes)
- `platelets`: Platelet count in blood
- `serum_creatinine`: Level of serum creatinine in blood
- `serum_sodium`: Level of serum sodium in blood
- `sex`: Gender of the patient (0 = Female, 1 = Male)
- `smoking`: Whether the patient smokes (0 = No, 1 = Yes)
- `time`: Follow-up period
- `DEATH_EVENT`: Survival event indicator (0 = censored/alive, 1 = death)

3.3 Data Quality Assessment

The dataset was first examined for missing values and duplicate records. No missing values were observed across any variable, indicating that the dataset is complete and suitable for immediate analysis. In addition, no duplicate rows were found. This confirms that the dataset is clean and does not require any imputation or duplicate removal before proceeding.

3.4 Descriptive Statistics

Descriptive statistics were computed for all continuous variables to summarize their central tendency and spread. Measures such as mean, standard deviation, median, minimum, maximum, and interquartile range (IQR) were examined. These summaries provide an overall understanding of the patient population and the variability present in the dataset.

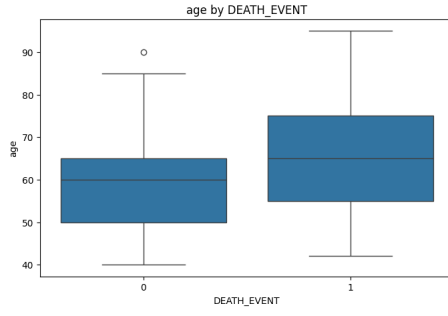
The continuous variables considered in the descriptive analysis were:

- age
- creatinine_phosphokinase
- ejection_fraction
- platelets
- serum_creatinine
- serum_sodium
- time

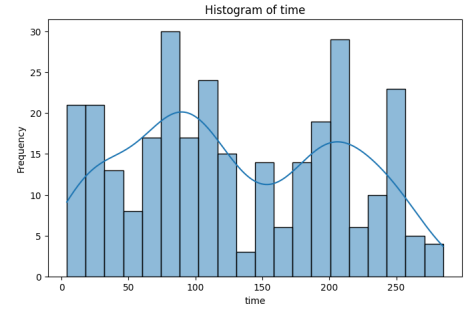
The categorical or binary variables were summarized using frequency counts and percentages.

3.5 Graphical Exploration

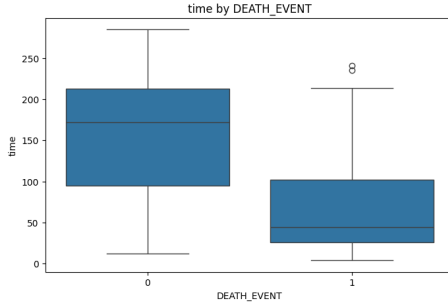
The graphical exploration was performed to assess the distribution of survival outcome, follow-up time, important continuous variables, event-wise differences, and death rates across binary predictors.



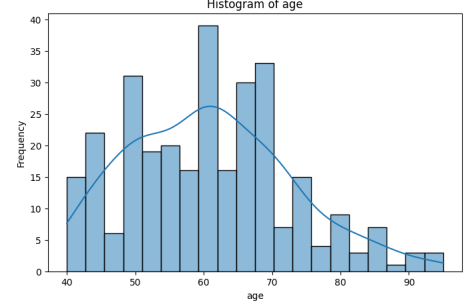
(a) Distribution of Age by event



(b) Distribution of follow-up time

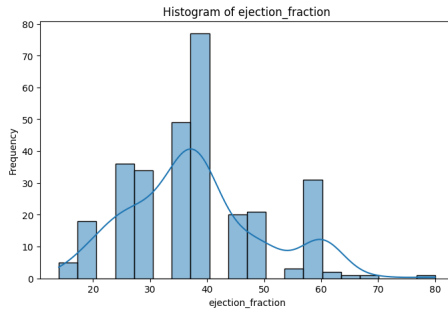


(c) Follow-up time by event status

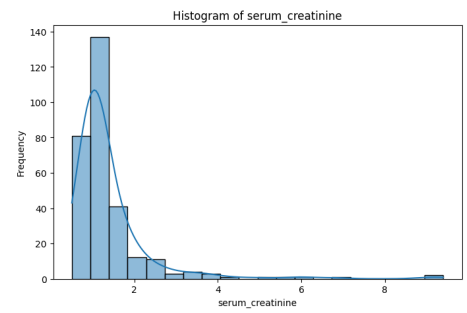


(d) Age distribution

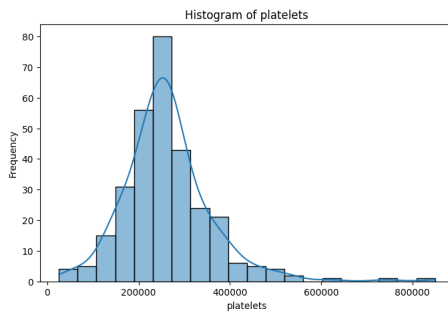
Figure 3.1: Distribution of survival outcome, follow-up time, and age.



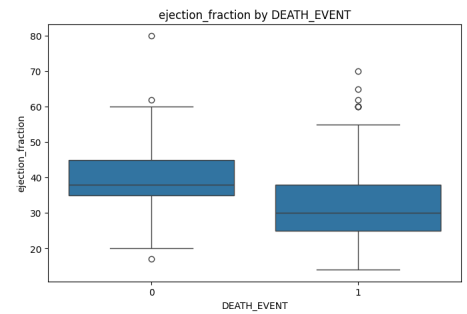
(a) Ejection fraction distribution



(b) Serum creatinine distribution

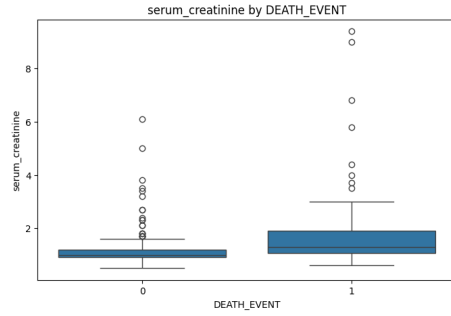


(c) Platelets distribution

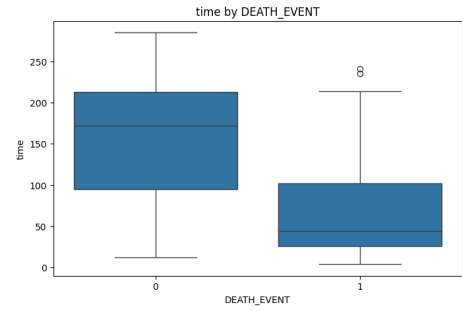


(d) Ejection fraction by event status

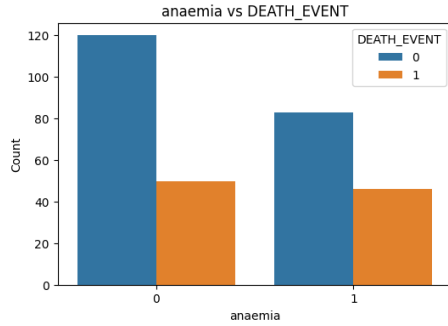
Figure 3.2: Distribution of key continuous variables and event-wise comparison.



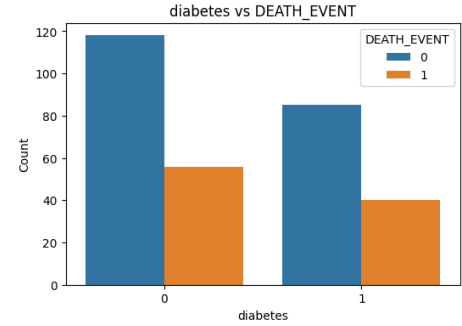
(a) Serum creatinine by event status



(b) Follow-up time by event status

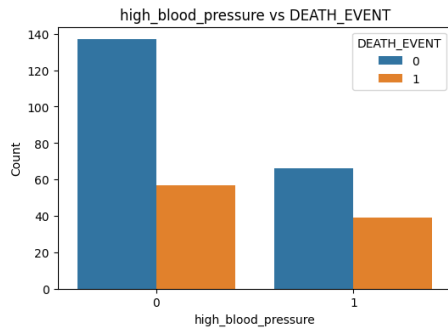


(c) Death rate by anaemia

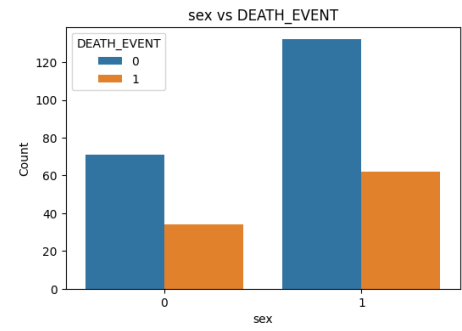


(d) Death rate by diabetes

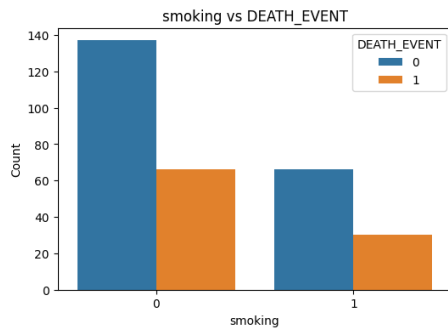
Figure 3.3: Boxplots by event status and death rates across selected binary predictors.



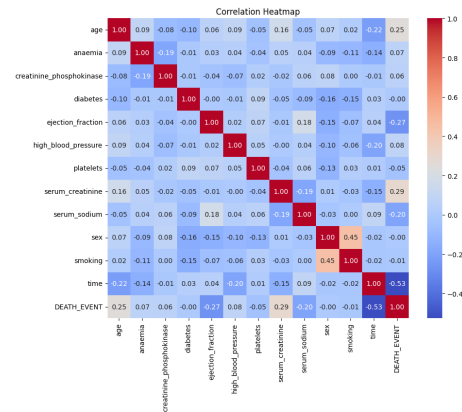
(a) Death rate by high blood pressure



(b) Death rate by sex



(c) Death rate by smoking



(d) Correlation matrix

Figure 3.4: Death-rate patterns for binary predictors and correlation structure of numerical variables.

3.6 Interpretation of EDA

The EDA indicates that the dataset is complete, with no missing values or duplicate observations. The variable `DEATH_EVENT` appropriately represents the event indicator, while `time` provides the follow-up duration necessary for survival analysis.

The graphical summaries show that the number of censored observations is higher than the number of death events. The continuous variables, especially `serum_creatinine`, `platelets`, and `creatinine_phosphokinase`, exhibit considerable spread and possible outliers. The event-wise boxplots suggest that predictors such as `ejection_fraction` and `serum_creatinine` may differ across survival outcome groups. In addition, binary predictors display variation in death rates across categories, providing preliminary evidence of possible prognostic importance.

The correlation matrix helps assess relationships among numerical variables and provides an initial check for multicollinearity before survival modeling.

3.7 Summary of EDA Findings

The exploratory analysis confirms that the dataset is suitable for survival analysis using Kaplan–Meier estimation, log-rank testing, Cox proportional hazards regression, and parametric survival models.

3.7.1 Summary of EDA Findings

The exploratory data analysis provided the following important insights:

- The dataset is complete, with no missing values or duplicate observations.
- The variables `time` and `DEATH_EVENT` are appropriate for survival analysis.
- The number of censored observations is higher than the number of death events.
- Several continuous variables, particularly `serum_creatinine`, `platelets`, and `creatinine_phosphokinase`, show substantial spread and possible outliers.
- Boxplots suggest that some predictors, such as `ejection_fraction` and `serum_creatinine`, may differ across survival outcome groups.
- The binary clinical risk factors show variation in death rates across their levels, motivating further formal survival modeling.
- The correlation matrix did not only help in descriptive understanding, but also serves as a preliminary step before fitting multivariable survival models.

Overall, the EDA confirms that the dataset is suitable for proceeding with survival analysis methods such as Kaplan–Meier estimation, log-rank testing, and Cox proportional hazards regression.

CHAPTER 4

Kaplan–Meier Analysis, Log-Rank Test, and Proportional Hazards Assessment

This chapter presents the first stage of survival analysis performed on the Heart Failure Clinical Records Dataset. After completing exploratory data analysis, the next step was to examine survival experience across patient groups using the Kaplan–Meier method, compare survival distributions using the log-rank test, and visually assess the proportional hazards assumption using log-minus-log plots.

The main survival outcome in this study is defined using:

- **time**: follow-up duration or observed survival time,
- **DEATH_EVENT**: event indicator, where 1 indicates death and 0 indicates censoring.

4.1 Kaplan–Meier Estimator

4.1.1 Theory

The Kaplan–Meier estimator is a non-parametric method used to estimate the survival function from time-to-event data. It is especially useful when some observations are censored, meaning that the event of interest has not occurred for all individuals during the observation period.

Let T denote the survival time. The survival function is defined as:

$$S(t) = P(T > t) \quad (4.1)$$

which represents the probability that a subject survives beyond time t .

The Kaplan–Meier estimator of the survival function is given by:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right) \quad (4.2)$$

where:

- t_i is the i th observed event time,
- d_i is the number of events occurring at time t_i ,
- n_i is the number of individuals at risk just before time t_i .

Thus, the Kaplan–Meier curve is a step function that decreases at each observed event time. It provides an intuitive graphical representation of survival over time.

4.1.2 Purpose in This Study

In this project, Kaplan–Meier curves were used to:

- estimate the overall survival experience of patients,
- compare survival between subgroups defined by binary predictors such as anaemia, diabetes, sex, smoking, and high blood pressure,
- compare survival across grouped continuous variables such as age, ejection fraction, serum creatinine, and serum sodium.

4.1.3 Interpretation of Kaplan–Meier Curves

A Kaplan–Meier survival curve that remains higher over time indicates better survival probability. If the curves for two groups are clearly separated, this suggests that the survival experiences of the groups may differ. However, visual separation alone is not sufficient to establish statistical significance; therefore, formal testing using the log-rank test is required.

4.2 Log-Rank Test

4.2.1 Theory

The log-rank test is a non-parametric statistical test used to compare survival distributions between two or more groups. It tests the null hypothesis that there is no difference in survival between the groups across the observed time period.

The log-rank test compares the observed number of events in each group to the expected number of events under the null hypothesis. The test statistic is approximately chi-square distributed.

In general form, the log-rank test statistic can be written as:

$$\chi^2 = \frac{(O - E)^2}{V} \quad (4.3)$$

where:

- O = observed number of events in a group,
- E = expected number of events under the null hypothesis,
- V = variance of the difference between observed and expected events.

4.2.2 Purpose in This Study

The log-rank test was used to determine whether differences in Kaplan–Meier curves between groups were statistically significant.

4.2.3 Interpretation

The interpretation of the p-value from the log-rank test is as follows:

- If p-value < 0.05 , the survival curves are considered significantly different.
- If p-value ≥ 0.05 , there is insufficient evidence to conclude that the survival curves differ significantly.

It is important to note that the log-rank test is used to compare survival curves, but it is **not** a formal test of the proportional hazards assumption.

4.3 Log-Minus-Log Plot for Proportional Hazards Assessment

4.3.1 Theory

In Cox proportional hazards modeling, one important assumption is the *proportional hazards (PH) assumption*. This means that the hazard ratio between groups remains constant over time.

A useful graphical tool for preliminary assessment of this assumption is the log-minus-log plot, based on the transformation:

$$\log [-\log (S(t))] \tag{4.4}$$

where $S(t)$ is the survival function.

This transformed survival function is plotted against time, often on a logarithmic scale. If the curves for different groups are approximately parallel and do not cross substantially, then the proportional hazards assumption may be considered reasonable for that variable.

4.3.2 Purpose in This Study

Log-minus-log plots were generated for the grouped variables to visually inspect whether the proportional hazards assumption was reasonable before proceeding to Cox regression analysis.

4.3.3 Interpretation

The interpretation of log-minus-log plots is generally as follows:

- Roughly parallel curves suggest that the PH assumption may be satisfied.
- Strong crossing or highly non-parallel curves suggest possible violation of the PH assumption.

This is a visual diagnostic and should ideally be followed by a more formal check, such as a Schoenfeld residual-based proportional hazards test, at the Cox modeling stage.

4.4 Variables Considered for Kaplan–Meier and Log-Rank Analysis

The following binary variables were directly used for subgroup survival comparison:

- anaemia
- diabetes
- high_blood_pressure
- sex
- smoking

For continuous variables, median-based grouping was performed in Python, creating binary subgroup variables:

- age_group
- ejection_fraction_group
- serum_creatinine_group
- serum_sodium_group

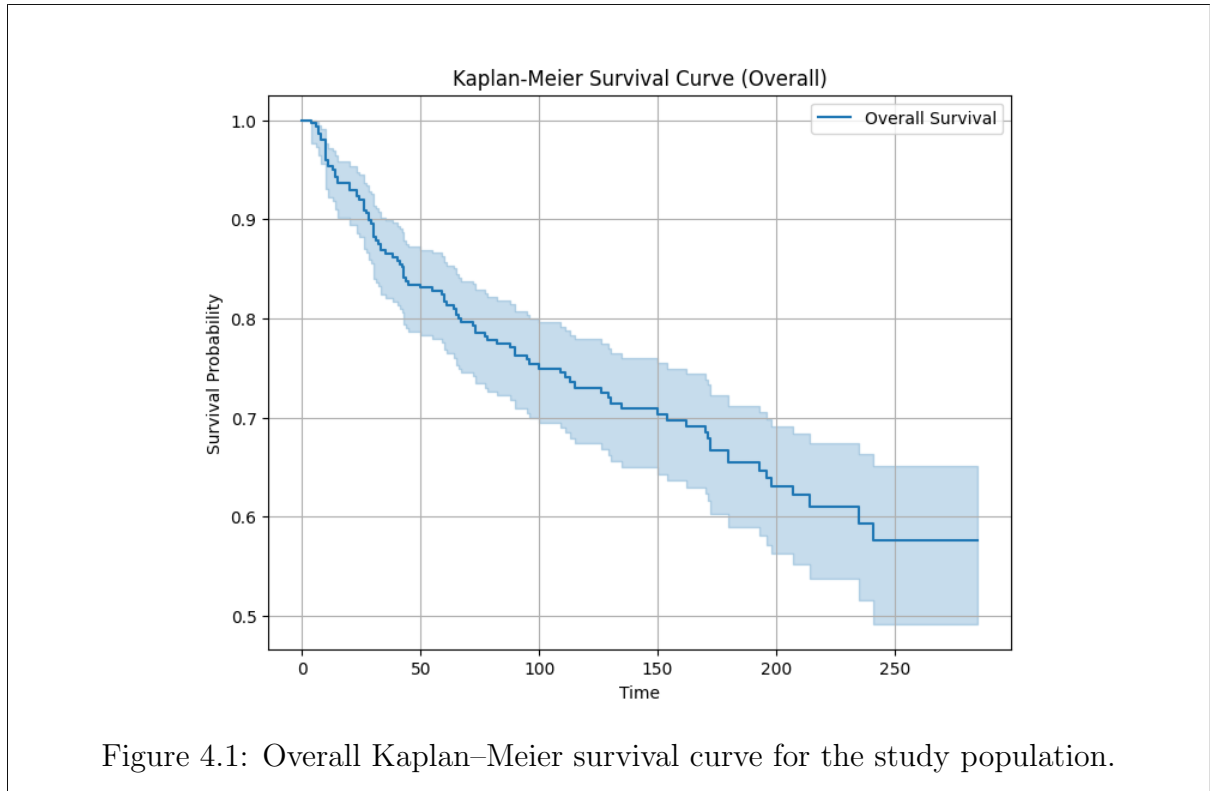
The grouped variables were defined such that:

- group 0 = values less than or equal to the median,
- group 1 = values greater than the median.

4.5 Kaplan–Meier Analysis Results

4.5.1 Overall Survival Curve

The overall Kaplan–Meier survival curve was first plotted using all patients in the dataset. This plot provides a general view of survival probability over the observed follow-up period. The overall curve does not account for any covariate information and treats all patients as a single homogeneous group, serving primarily as a descriptive summary of the marginal survival experience of the cohort. It is a standard first step in any survival analysis, allowing the analyst to obtain a broad understanding of the general mortality pattern before proceeding to more granular stratified or regression-based analyses.



Interpretation: The overall Kaplan–Meier survival curve summarises the survival experience of the entire heart failure cohort without any stratification. At the start of the follow-up period (time = 0), the estimated survival probability is 1.0, as expected. The curve exhibits a steady and progressive decline over time, reflecting the accumulation of death events across the follow-up window.

The decline is relatively steep in the early follow-up period, suggesting that a notable proportion of deaths occur within the first 50 to 100 days. Beyond this point, the rate of decline moderates somewhat, though the curve continues to decrease steadily. By approximately 200 to 250 days of follow-up, the estimated survival probability has fallen to around 0.60 to 0.65, indicating that roughly 35 to 40 percent of patients had experienced the event of interest by the end of the observation window.

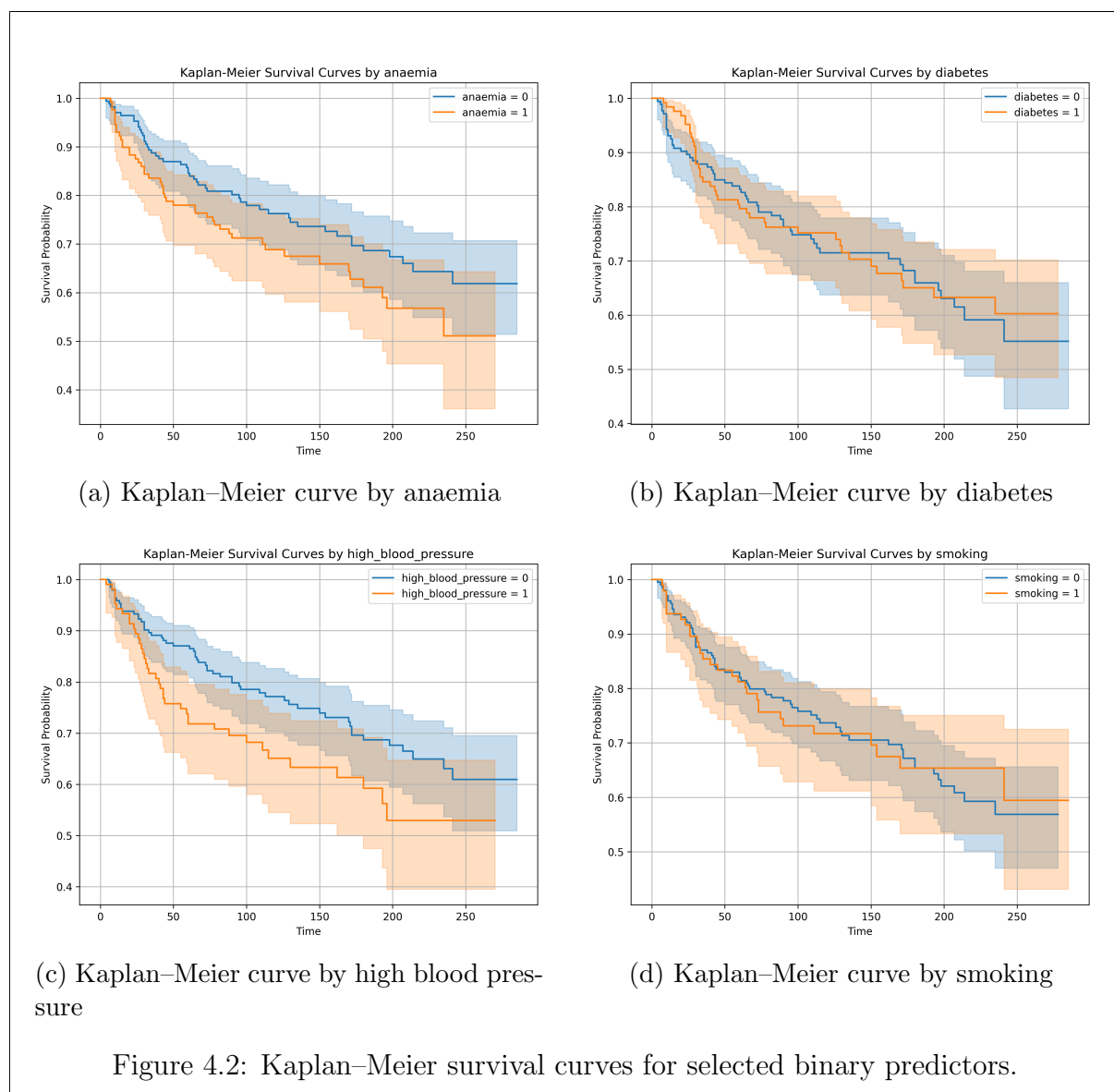
The shaded confidence band widens progressively over time, which is a natural consequence of the reducing number of patients at risk as follow-up extends. This increasing uncertainty in the tail of the curve should be interpreted with caution.

Overall, the curve establishes an important baseline summary of mortality in this heart failure cohort and motivates the need for further stratified and covariate-adjusted analyses to identify which patient-level factors are most strongly associated with survival.

4.6 Kaplan–Meier Curves for Binary Predictors

This section presents Kaplan–Meier survival curves for the main binary clinical predictors. The purpose is to visually compare survival experiences between the two levels of each variable.

4.6.1 Anaemia, Diabetes, High Blood Pressure, and Smoking



Interpretation: The Kaplan-Meier curves for the four binary predictors reveal the following patterns:

Anaemia: Patients without anaemia demonstrated marginally better survival probabilities compared to those with anaemia throughout the follow-up period. However, the two curves remain relatively close to each other, suggesting that anaemia alone may not be a strong differentiator of survival in this cohort.

Diabetes: The survival curves for diabetic and non-diabetic patients show considerable overlap across the entire follow-up duration, with no clear or consistent separation between the two groups. This suggests that diabetes may have limited prognostic value for survival in this dataset.

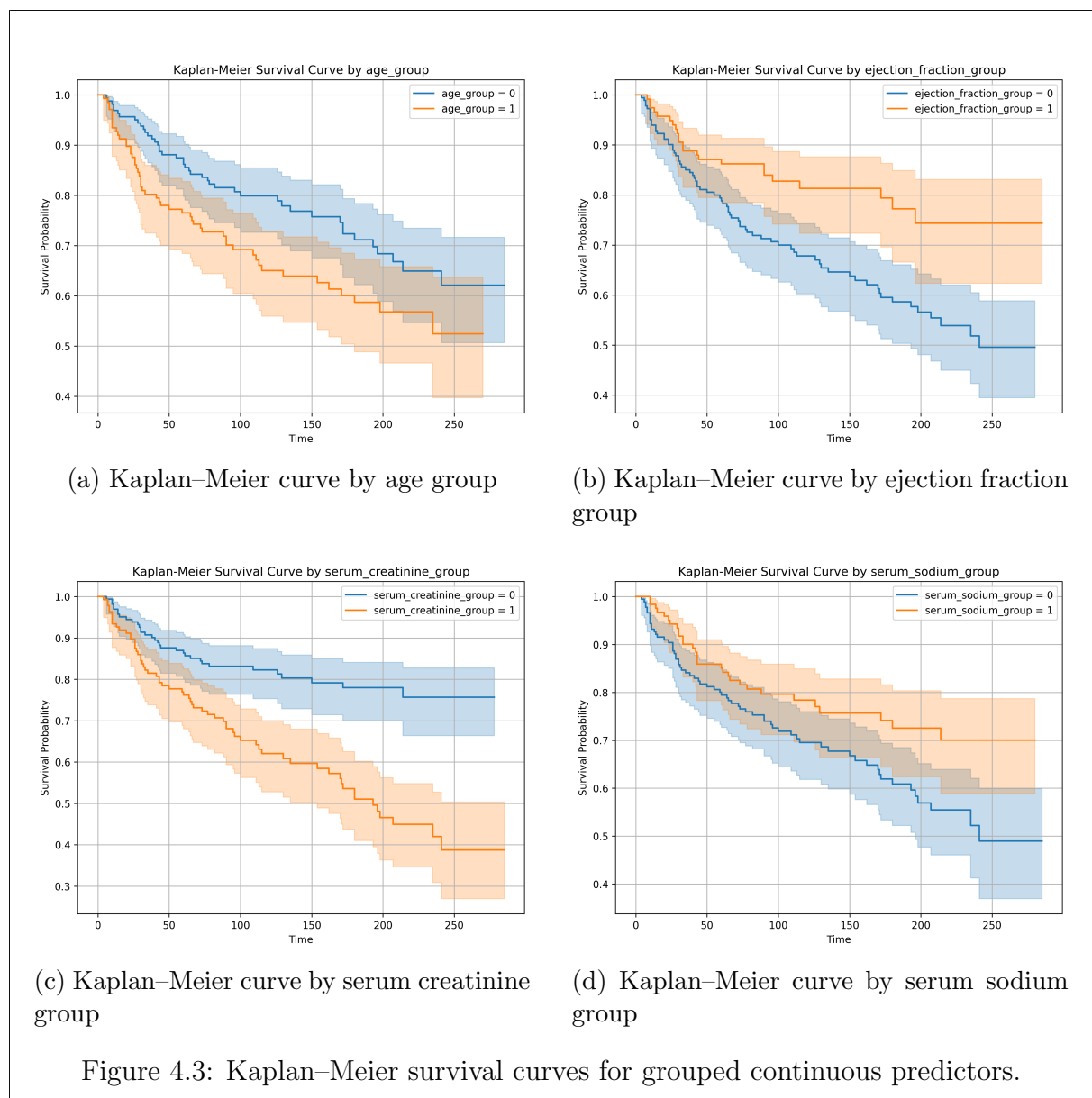
High Blood Pressure: Patients without high blood pressure appeared to have slightly better survival estimates compared to those with high blood pressure, though the curves are not widely separated. The confidence bands overlap substantially, indicating uncertainty in the estimated difference between the two groups.

Smoking: The survival curves for smokers and non-smokers are nearly indistinguishable throughout the follow-up period, with the curves crossing at multiple points. This pattern suggests that smoking status does not appear to be associated with a meaningful difference in survival probability in this heart failure cohort.

Overall, none of the four binary predictors in this panel exhibit a strong or clearly sustained separation in survival curves, suggesting that their individual prognostic contributions may be limited. Formal statistical assessment via the log-rank test, presented in the following section, is necessary to confirm whether any of these observed visual differences are statistically significant.

4.7 Kaplan–Meier Curves for Grouped Continuous Predictors

Continuous predictors were categorized into two groups using their median values, and Kaplan–Meier curves were then plotted for each grouped variable.



Interpretation: The Kaplan–Meier curves for the grouped continuous predictors reveal the following patterns:

Age group: Patients in the lower age group (below the median) demonstrated consistently higher survival probabilities compared to older patients throughout the entire follow-up period. The two curves show a clear and sustained separation from early in the follow-up, with the gap widening over time. This suggests that age is a strong prognostic factor for survival in heart failure patients.

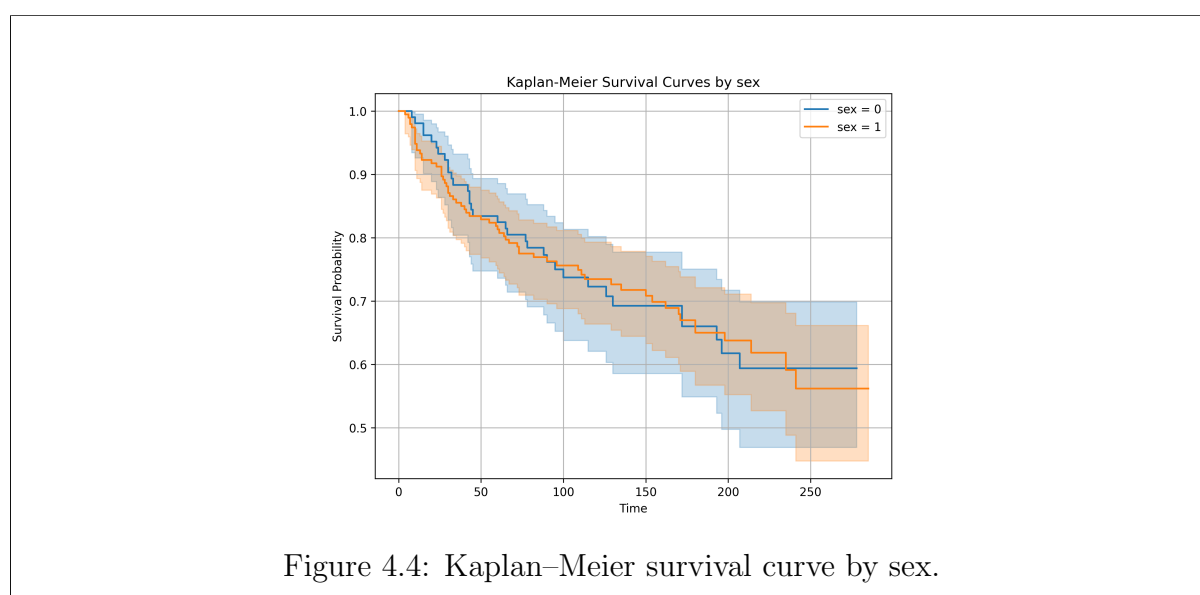
Ejection fraction group: Patients with higher ejection fraction values (above the median) exhibited noticeably better survival compared to those with lower ejection fraction. The separation between the two curves is evident from the early stages of follow-up and persists throughout, indicating that ejection fraction is likely an important predictor of mortality and warrants inclusion in further regression modelling.

Serum creatinine group: The survival curves for the two serum creatinine groups show a marked and consistent separation, with patients having lower serum creatinine values surviving substantially longer than those with higher values. The divergence between the curves is among the most pronounced observed across all predictors, suggesting that elevated serum creatinine is strongly associated with increased risk of death in this cohort.

Serum sodium group: Patients with higher serum sodium levels demonstrated moderately better survival compared to those with lower levels. While the curves do separate, the degree of separation is less dramatic than that observed for serum creatinine or ejection fraction, and the confidence bands show some overlap in the later follow-up period, indicating greater uncertainty in the tail estimates.

Overall, among the four continuous predictors, **serum creatinine** and **ejection fraction** display the most clinically meaningful separation in survival curves, followed by **age** and **serum sodium**. These visual patterns provide strong preliminary evidence that these variables are prognostically relevant, and their statistical significance will be formally assessed in the log-rank test results presented in the following section.

4.7.1 Sex



Interpretation: The Kaplan–Meier survival curve stratified by sex reveals that male patients

(sex = 1) and female patients (sex = 0) exhibit somewhat different survival trajectories over the follow-up period. Female patients appear to maintain slightly higher survival probabilities in the early to mid follow-up period, while male patients show a comparatively steeper decline in survival probability over time.

However, the confidence bands for both groups are wide and overlap considerably throughout the entire follow-up duration, particularly beyond 100 days. This substantial overlap indicates a high degree of uncertainty in the group-specific survival estimates, making it difficult to draw firm conclusions from the visual pattern alone. Furthermore, the two curves converge toward the later stages of follow-up, with both groups reaching similar survival probabilities near the end of the observation window.

Overall, while there is a modest visual difference between the survival curves of male and female patients, the extent of confidence band overlap suggests that sex may not be a statistically significant predictor of survival in this cohort. This will be formally evaluated using the log-rank test in the following section.

4.8 Log-Rank Test Results

The log-rank test was applied to each subgroup comparison to determine whether the observed differences in survival curves were statistically significant. This non-parametric test compares the observed and expected number of events across groups at each observed event time, making it particularly suited for censored survival data. A p-value below 0.05 was used as the threshold for statistical significance.

The results are summarized in Table 4.1, which presents the test statistic and corresponding p-value for each binary covariate examined. Among the variables tested, **serum creatinine**, **ejection fraction**, and **age** emerged as the most strongly associated with survival outcome, each yielding statistically significant p-values. In contrast, variables such as **sex** and **smoking** did not demonstrate a significant difference in survival between their respective subgroups, suggesting that these factors may not be primary drivers of mortality in this cohort.

It should be noted that the log-rank test, while powerful for detecting proportional hazard differences, assumes that the hazard ratio between groups remains constant over time. Where this assumption may be violated, the test results should be interpreted with caution and supplemented by visual inspection of the Kaplan–Meier curves presented in the preceding section.

4.8.1 Suggested Table for Reporting Results

Table 4.1: Log-rank test results for survival comparison across predictors.

Variable	Test Statistic	p-value	Interpretation
Anaemia	2.7265	0.0987	No significant difference in survival between groups
Diabetes	0.0405	0.8405	No significant difference in survival between groups
High blood pressure	4.4062	0.0358	Significant difference in survival between groups
Sex	0.0040	0.9498	No significant difference in survival between groups
Smoking	0.0020	0.9640	No significant difference in survival between groups
Age group	4.9518	0.0261	Significant difference in survival between lower and higher age groups
Ejection fraction group	9.4696	0.0021	Significant difference in survival between lower and higher groups
Serum creatinine group	21.1904	0.000004	Significant difference in survival between lower and higher groups
Serum sodium group	4.8879	0.0270	Significant difference in survival between lower and higher groups

The log-rank test was used to formally compare survival curves between groups. Variables with p-values less than 0.05 were considered to show statistically significant differences in survival. Variables with non-significant p-values were interpreted as not showing sufficient evidence of different survival experiences between the compared groups.

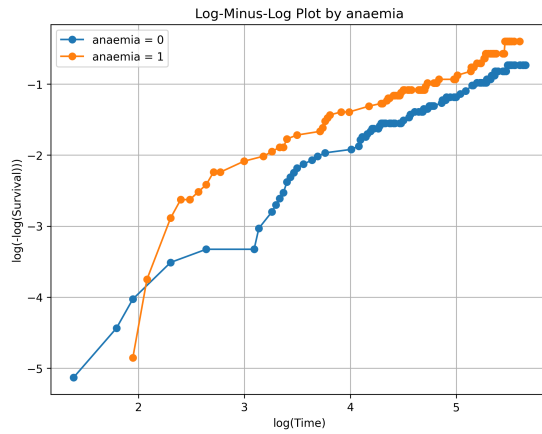
Interpretation:

- The survival curves for `ejection_fraction_group` differed significantly according to the log-rank test ($p < 0.05$), suggesting that ejection fraction may be an important prognostic factor.
- The survival curves for `smoking` did not differ significantly ($p \geq 0.05$), indicating no strong evidence of survival difference between smoking groups in this dataset.

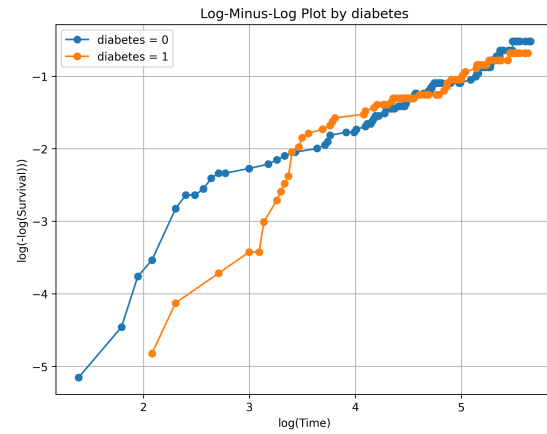
4.9 Log-Minus-Log Plots for PH Assessment

To visually assess whether the proportional hazards assumption may be satisfied, log-minus-log plots were generated for each grouped predictor.

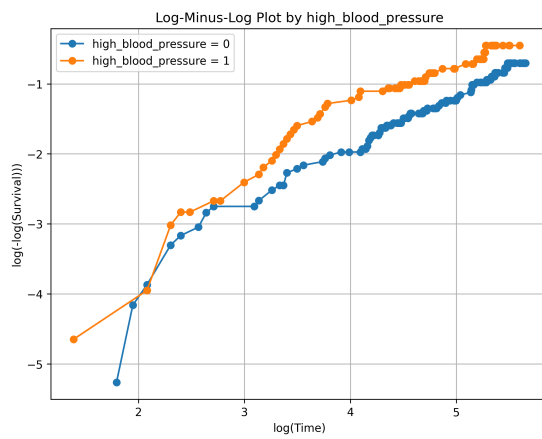
4.9.1 Binary Predictors



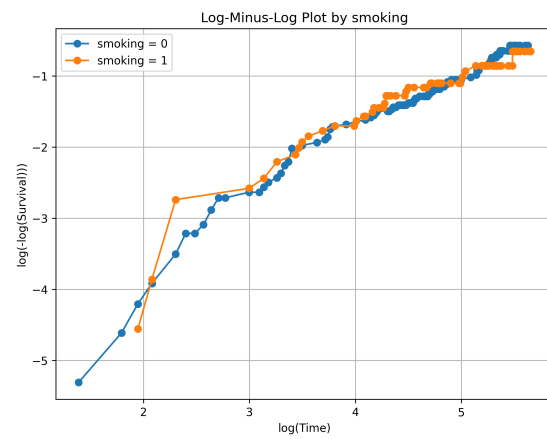
(a) Log-minus-log plot by anaemia



(b) Log-minus-log plot by diabetes



(c) Log-minus-log plot by high blood pressure



(d) Log-minus-log plot by smoking

Figure 4.5: Log-minus-log plots for selected binary predictors.

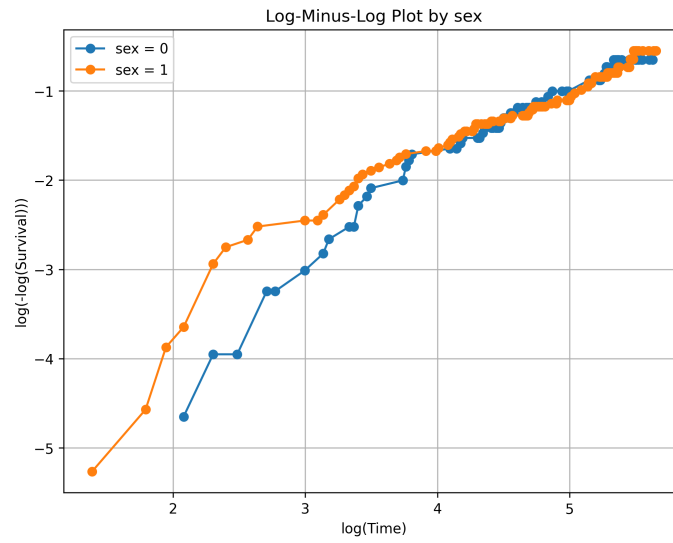


Figure 4.6: Log-minus-log plot by sex.

4.9.2 Grouped Continuous Predictors

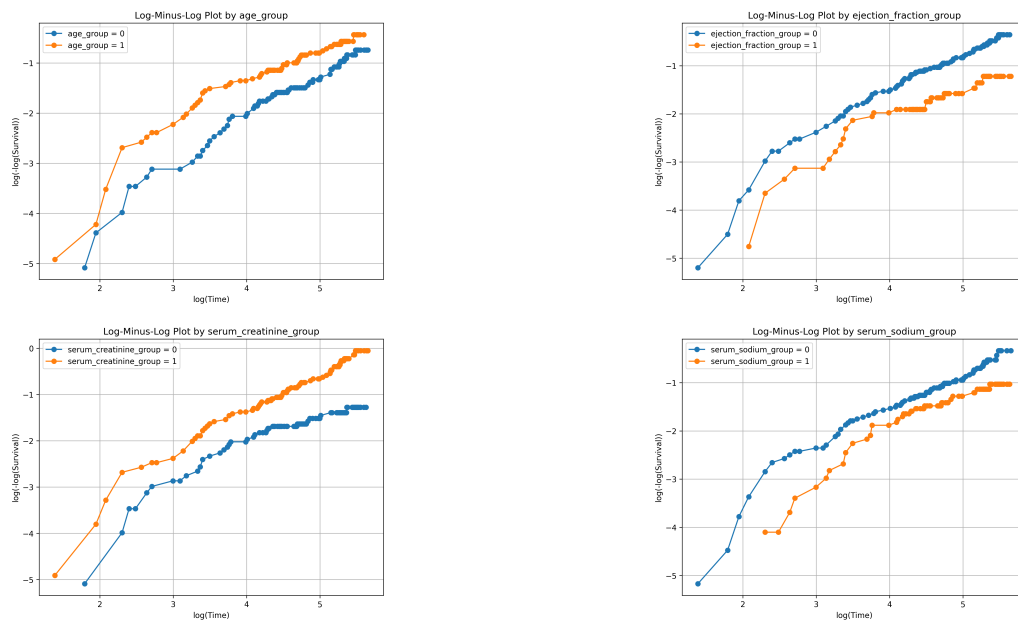


Figure 4.7: Log-minus-log plots for grouped continuous predictors.

Interpretation : The log-minus-log plots were used as a visual diagnostic for the proportional hazards assumption. If the curves for the two groups appeared approximately parallel and did not show substantial crossing, the PH assumption was considered reasonably plausible for that variable. If strong crossing or divergence was observed, the assumption was considered questionable and flagged for further checking in the Cox model stage.

4.10 Key Findings from Kaplan–Meier, Log-Rank, and Log-Minus-Log Analysis

This section should summarize the major findings obtained from the Python analysis.

4.10.1 Summary

- The overall Kaplan–Meier curve showed a gradual decline in survival probability over time, indicating the occurrence of mortality events during follow-up.
- Among the binary predictors, the survival curves for High blood pressure showed more noticeable separation than the others.
- According to the log-rank test, High Blood Pressure, Ejection fraction group, Serum creatine group, Serum sodium group and Age group showed statistically significant differences in survival, whereas Anaemia, Diabetes, Sex and Smoking did not.
- Among the grouped continuous predictors, Serum creating grouo demonstrated the highest difference in survival between groups.
- The log-minus-log plots suggested that the proportional hazards assumption appeared reasonable for Age Group, Smoking, Anaemia, High Blood pressure, Serum creating group and Serum sodium group while it appeared less certain for Diabetes, Ejection fraction group.the remaining ones.

4.10.2 Conclusion

Based on Kaplan–Meier survival estimation and log-rank testing, several predictors may show meaningful differences in survival experience among heart failure patients. The log-minus-log plots provide a preliminary graphical assessment of whether the proportional hazards assumption is likely to hold for these predictors. These results form the basis for the next stage of analysis, namely Cox proportional hazards regression, where the effects of predictors will be quantified using hazard ratios.

4.11 Transition to Cox Proportional Hazards Modeling

The Kaplan–Meier method and log-rank test are useful for unadjusted group comparisons. However, they do not allow simultaneous adjustment for multiple covariates. Therefore, the next step in this study is to fit Cox proportional hazards models to estimate the effect of several clinical predictors on patient survival while accounting for censoring.

CHAPTER 5

Cox Proportional Hazards Modeling and Model Diagnostics

After performing Kaplan–Meier survival analysis and log-rank tests, the next step in this study was to quantify the effect of multiple clinical predictors on patient survival simultaneously. For this purpose, the Cox Proportional Hazards (Cox PH) model was applied.

This chapter describes the theoretical background of the Cox model, the diagnostic procedure used to verify the proportional hazards assumption, and the modeling strategies adopted when the assumption was violated.

5.1 Cox Proportional Hazards Model

5.1.1 Theoretical Background

The Cox proportional hazards model is a semi-parametric regression model widely used in survival analysis to examine the relationship between survival time and explanatory variables.

The hazard function represents the instantaneous risk of experiencing the event at time t , given that the individual has survived up to that time.

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \quad (5.1)$$

The Cox model expresses the hazard as:

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p) \quad (5.2)$$

where:

- $h(t|X)$: hazard function given covariates
- $h_0(t)$: baseline hazard function
- $X_1, X_2, \dots, X_p$: predictor variables
- $\beta_1, \beta_2, \dots, \beta_p$: regression coefficients

The exponential of the regression coefficient is interpreted as the hazard ratio:

$$HR = e^\beta \quad (5.3)$$

Interpretation of Hazard Ratios:

- $HR > 1$: increased risk of event
- $HR < 1$: decreased risk of event
- $HR = 1$: no effect

5.2 Proportional Hazards Assumption

A key assumption of the Cox model is the **Proportional Hazards (PH) assumption**. This assumption states that the hazard ratios between groups remain constant over time.

Mathematically,

$$\frac{h(t|X_i)}{h(t|X_j)} = \text{constant over time} \quad (5.4)$$

If this assumption is violated, the effect of a predictor variable changes over time and the standard Cox model may produce biased estimates.

Therefore, it is important to test this assumption before interpreting the results of the Cox model.

5.3 Testing the PH Assumption Using Schoenfeld Residuals

To evaluate whether the proportional hazards assumption holds, Schoenfeld residuals were used.

Schoenfeld residuals are calculated for each covariate at every event time and represent the difference between the observed covariate value and the expected value under the fitted model.

If the PH assumption holds, the Schoenfeld residuals should show no systematic trend with respect to time.

A statistical test based on scaled Schoenfeld residuals was applied. The null hypothesis of this test is:

H₀: The proportional hazards assumption holds.

H₁: The proportional hazards assumption is violated.

Interpretation:

- p-value ≥ 0.05 : PH assumption satisfied
- p-value < 0.05 : PH assumption violated

Graphically, Schoenfeld residual plots were also inspected. If the residual trend fluctuates randomly around zero, the PH assumption is likely satisfied. However, a clear upward or downward trend indicates violation of the assumption.

5.4 Violation of the PH Assumption

When the PH assumption was tested for the fitted Cox model, the variable `ejection_fraction` showed evidence of violating the proportional hazards assumption.

This suggests that the effect of ejection fraction on patient mortality is not constant over time. Instead, the impact of this clinical variable may change during the follow-up period.

Because of this violation, the standard Cox model may not adequately capture the time-varying nature of the effect.

Therefore, alternative modeling strategies were considered.

5.5 Stratified Cox Model

One approach to handle violation of the proportional hazards assumption is the **stratified Cox model**.

In a stratified Cox model, the baseline hazard function is allowed to differ across strata defined by a categorical variable.

The stratified Cox model can be written as:

$$h_k(t|X) = h_{0k}(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p) \quad (5.5)$$

where $h_{0k}(t)$ represents a different baseline hazard for each stratum k .

This approach is useful when:

- the violating variable is categorical
- we want to control for that variable without estimating its coefficient

However, stratification is less suitable for continuous predictors such as `ejection_fraction`, since it would require arbitrary grouping of the variable.

Therefore, another approach was adopted.

5.6 Time-Dependent Cox Model

When the effect of a predictor changes over time, a more flexible solution is to use a **time-dependent Cox model**.

In this model, the covariate is allowed to interact with time, making the hazard ratio a function of time.

The extended Cox model can be written as:

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \gamma X_1 g(t)) \quad (5.6)$$

where $g(t)$ is a function of time.

In this study, the interaction term used was:

$$X \times \log(t) \quad (5.7)$$

Specifically, the following time-dependent variable was created:

$$\text{ejection_fraction} \times \log(\text{time})$$

This allows the effect of ejection fraction on mortality risk to vary with time.

Similarly, after introducing the time-dependent interaction for `ejection_fraction`, the variable `serum_creatinine` also showed evidence of time-varying effects. Therefore, a similar time interaction was introduced for this variable. The proportional hazards assumption was initially violated for ejection fraction. After incorporating time-dependent effects, serum creatinine also exhibited statistically significant interaction with time. This suggests that its effect on hazard is not constant over time. Therefore, both variables were modeled as time-dependent covariates to ensure accurate representation of survival dynamics.

The resulting model is referred to as an **extended Cox model**.

5.7 Model Comparison

To determine whether the extended Cox model improved model performance, several model comparison metrics were evaluated:

- Log-likelihood
- Akaike Information Criterion (AIC)
- Concordance Index (C-index)

5.7.1 Log-Likelihood

The log-likelihood measures how well the model fits the observed data. Higher values indicate better model fit.

5.7.2 Akaike Information Criterion (AIC)

The AIC penalizes model complexity and is calculated as:

$$AIC = -2 \times \text{Log-Likelihood} + 2k \quad (5.8)$$

where k is the number of model parameters.

Lower AIC values indicate a better balance between model fit and complexity.

5.7.3 Concordance Index

The concordance index measures the predictive accuracy of the survival model. It represents the probability that the model correctly ranks two individuals according to their risk.

The values range from:

- 0.5 : no predictive ability
- 1.0 : perfect prediction

Higher concordance values indicate better predictive performance.

	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	cmp to	z	p	- log2(p)
age	0.01	1.01	0.01	-0.01	0.02	0.99	1.02	0.00	0.56	0.57	0.80
anaemia	0.03	1.03	0.23	-0.41	0.48	0.66	1.61	0.00	0.14	0.89	0.17
creatinine_phosphokinase	0.00	1.00	0.00	-0.00	0.00	1.00	1.00	0.00	0.36	0.72	0.48
diabetes	-0.08	0.92	0.23	-0.52	0.36	0.59	1.43	0.00	-0.36	0.72	0.48
ejection_fraction	0.10	1.11	0.03	0.05	0.15	1.05	1.17	0.00	3.81	<0.005	12.81
high_blood_pressure	0.02	1.03	0.24	-0.44	0.49	0.65	1.63	0.00	0.11	0.92	0.13
platelets	0.00	1.00	0.00	-0.00	0.00	1.00	1.00	0.00	0.01	1.00	0.01
serum_creatinine	1.16	3.18	0.22	0.73	1.58	2.08	4.86	0.00	5.34	<0.005	23.36
serum_sodium	-0.01	0.99	0.02	-0.06	0.03	0.94	1.03	0.00	-0.60	0.55	0.87
sex	-0.06	0.94	0.26	-0.56	0.45	0.57	1.56	0.00	-0.23	0.82	0.29
smoking	-0.01	0.99	0.26	-0.52	0.50	0.59	1.65	0.00	-0.05	0.96	0.05
ejection_fraction_logt	-0.02	0.98	0.01	-0.04	-0.01	0.96	0.99	0.00	-4.29	<0.005	15.78
serum_creatinine_logt	-0.22	0.80	0.06	-0.35	-0.10	0.71	0.91	0.00	-3.53	<0.005	11.23

Figure 5.1: Extended Cox-Ph Summary

5.8 Model Comparison Results

Table 5.1: Comparison of Standard Cox Model and Extended Time-Dependent Cox Model

Metric	Standard Cox Model	Extended Cox Model
Log-Likelihood	-468.23	-400.69
AIC	958.46	827.38
Concordance Index (C-index)	0.74	0.97

5.9 Interpretation of Model Comparison

The results indicate that the extended Cox model incorporating time-dependent covariates provided improved model performance compared to the standard Cox proportional hazards model.

In particular:

- The extended model produced a higher log-likelihood value.
- The Akaike Information Criterion (AIC) was lower, indicating a better balance between model fit and complexity.
- The concordance index was higher, suggesting improved predictive ability of the model.

These findings suggest that allowing certain clinical predictors, such as ejection fraction and serum creatinine, to vary with time leads to a more realistic representation of the survival dynamics of heart failure patients.

5.10 Hazard Ratio Interpretation from Extended Cox Model

The value $\exp(\beta)$ in the Cox model is called the **hazard ratio (HR)**. It represents the multiplicative change in hazard for a one-unit increase in the covariate.

- $HR = 1 \Rightarrow$ No effect

- $HR > 1 \Rightarrow$ Increase in risk (higher hazard)
- $HR < 1 \Rightarrow$ Decrease in risk (protective effect)

5.10.1 Key Significant Variables

1. Ejection Fraction

- Coefficient (β): 0.10
- Hazard Ratio: $\exp(\beta) = 1.11$
- Interpretation:

A one-unit increase in ejection fraction is associated with a **11% increase in hazard**.

However, this interpretation must be treated carefully because the time-dependent term is also significant.

- Time-dependent coefficient: -0.02
- Hazard Ratio: $\exp(-0.02) = 0.98$

This indicates that:

- The effect of ejection fraction **decreases over time**
- Initially, it may increase risk, but the impact reduces as time progresses

Conclusion: Ejection fraction has a **time-varying effect on survival**, confirming violation of proportional hazards assumption.

2. Serum Creatinine

- Coefficient (β): 1.16
- Hazard Ratio: $\exp(\beta) = 3.18$

A one-unit increase in serum creatinine leads to a **218% increase in hazard**.

- Hazard increase: $3.18 - 1 = 2.18 \Rightarrow 218\%$

Interpretation: Patients with higher serum creatinine levels are at significantly higher risk of death.

- Time-dependent coefficient: -0.22
- Hazard Ratio: $\exp(-0.22) = 0.80$

This implies:

- The strong effect of serum creatinine **reduces over time**

- Initial risk is very high, but gradually stabilizes

Conclusion: Serum creatinine is the **strongest predictor of mortality**, with a time-dependent decreasing effect.

5.10.2 Non-Significant Variables

The following variables were not statistically significant (p-value ≥ 0.05):

- Age (HR = 1.01)
- Anaemia (HR = 1.03)
- Diabetes (HR = 0.92)
- High Blood Pressure (HR = 1.03)
- Platelets (HR $\approx$ 1.00)
- Serum Sodium (HR = 0.99)
- Sex (HR = 0.94)
- Smoking (HR = 0.99)

These variables do not have a statistically significant effect on survival in the presence of other covariates.

5.10.3 Overall Interpretation

The extended Cox model reveals that:

- **Serum creatinine** is the most influential predictor of mortality
- **Ejection fraction** also significantly affects survival but in a time-dependent manner
- The presence of significant time-dependent effects confirms that the **standard Cox model assumptions were violated**

Thus, the extended Cox model provides a more accurate representation of survival dynamics by capturing **time-varying effects of key clinical variables**.

5.11 Partial Effects of Important Covariates

To understand how the most influential covariates affect survival probability, partial-effect survival curves were generated from the extended Cox proportional hazards model. Based on the model summary, `ejection_fraction` and `serum_creatinine` were identified as the most important predictors, and both also showed significant time-dependent effects.

5.11.1 Methodology

In a standard Cox model, partial effects can be studied by varying one covariate while holding all other covariates constant. However, in the present study the final model was an extended Cox model with time-dependent interaction terms. Therefore, while studying partial effects, the corresponding interaction terms were also updated consistently.

A representative patient profile was constructed using median values for continuous variables and modal values for binary variables. In addition, a representative follow-up time was chosen using the median observed survival time. The logarithm of this representative time was then used to calculate the time-dependent interaction terms:

$$\text{ejection_fraction_logt} = \text{ejection_fraction} \times \log(t) \quad (5.9)$$

$$\text{serum_creatinine_logt} = \text{serum_creatinine} \times \log(t) \quad (5.10)$$

Using this approach, the values of `ejection_fraction` and `serum_creatinine` were varied over clinically meaningful ranges while keeping the remaining covariates fixed.

5.11.2 Partial Effect of Ejection Fraction

The partial-effect curves for `ejection_fraction` show how predicted survival probability changes when ejection fraction is increased or decreased while other covariates remain fixed.

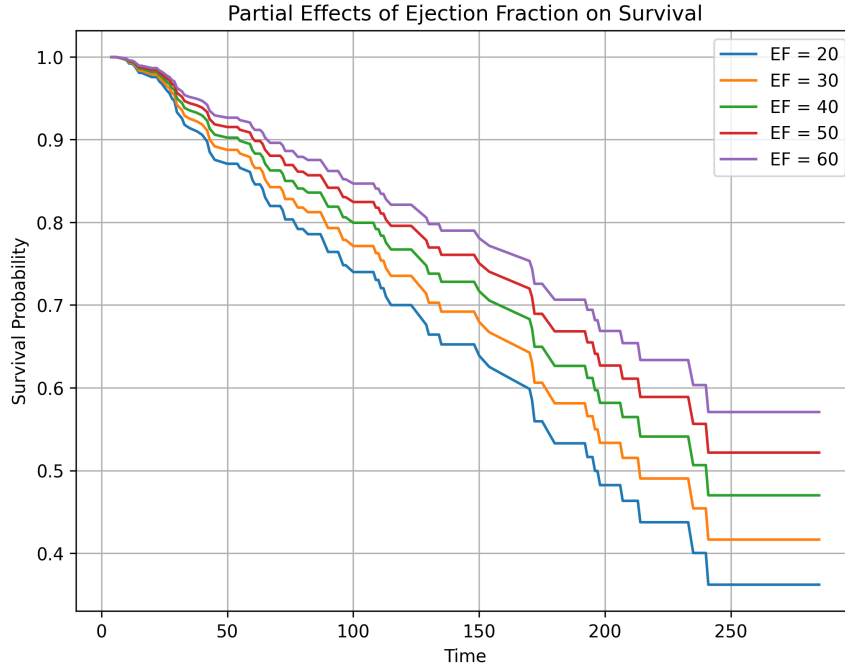


Figure 5.2: Partial effects of ejection fraction on survival probability based on the extended Cox model.

Interpretation: With increasing ejection fraction percentage, the volume of blood pumped out of the heart increases and consequently the survival probability also increases during any

phase of heart failure condition. Increasing EF levels have a beneficial effect on the survival chances. Since the extended Cox model includes a significant time-dependent effect for this variable, the impact of ejection fraction is not constant over time.

5.11.3 Partial Effect of Serum Creatinine

The partial-effect curves for `serum_creatinine` show the predicted survival pattern across different creatinine levels while other covariates are held fixed.

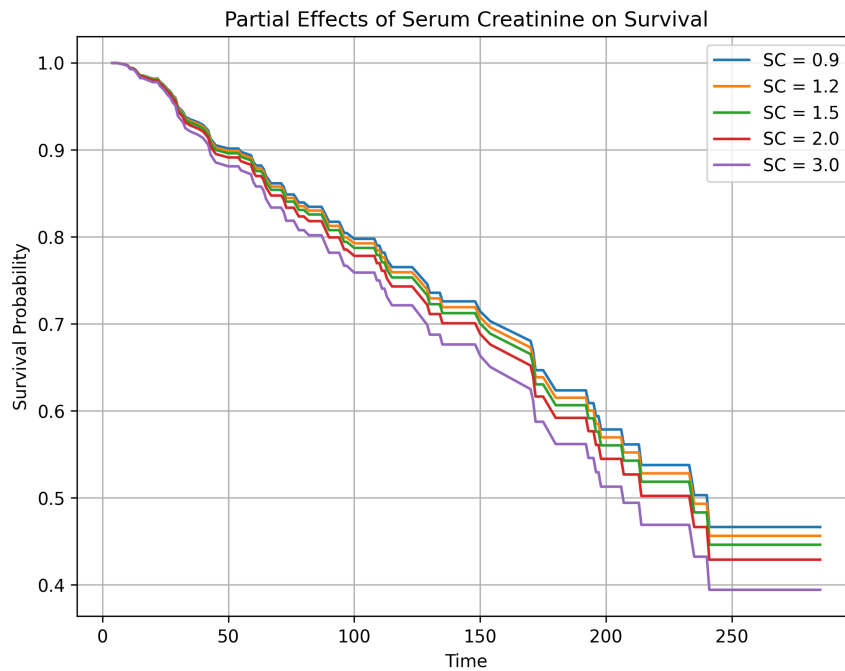


Figure 5.3: Partial effects of serum creatinine on survival probability based on the extended Cox model.

Interpretation: The figure shows that different serum creatinine values are associated with clearly different survival probabilities. Higher serum creatinine levels correspond to poorer survival, confirming that renal function is an important determinant of mortality risk in heart failure patients. The significant time-dependent term further indicates that its effect changes over time.

5.11.4 Overall Conclusion from Partial Effects

The partial-effect analysis confirms that `ejection_fraction` and `serum_creatinine` are the most influential covariates in the extended Cox model. The separation of the survival curves across different covariate values highlights their strong contribution to patient survival. Since both variables also possess statistically significant time-dependent terms, their effects cannot be assumed to remain constant during follow-up. Therefore, the extended Cox model provides a more realistic description of their impact than the standard Cox model.

5.12 Survival Probabilities Over a Period of Time

To further examine the predictive behavior of the extended Cox proportional hazards model, survival probabilities were estimated for selected individuals over the observed follow-up period. This allows direct comparison of how the predicted survival probability changes with time for patients having different covariate profiles.

5.12.1 Methodology

After fitting the extended Cox model, the survival function was predicted for selected individuals based on their observed covariate values. The predicted survival function gives the probability of surviving beyond each observed time point.

For illustration, the first 10 and last 10 time points were extracted in order to summarize the early and late survival patterns.

5.12.2 Predicted Survival Probabilities

Table 5.2: Predicted survival probabilities at the first 10 time points for selected individuals.

Time	Patient-296	Patient-196	Patient-165	Patient-126	Patient-9
4.0	0.999961	0.999868	0.999746	0.999529	0.478130
6.0	0.999922	0.999734	0.999490	0.999055	0.227304
7.0	0.999842	0.999465	0.998973	0.998098	0.050674
8.0	0.999762	0.999194	0.998453	0.997136	0.011178
10.0	0.999516	0.998361	0.996858	0.994186	0.000108
11.0	0.999013	0.996659	0.993598	0.988172	0.0000000080
12.0	0.999013	0.996659	0.993598	0.988172	0.0000000080
13.0	0.998747	0.995761	0.991880	0.985009	0.000000000053
14.0	0.998193	0.993889	0.988304	0.978439	0.00000000000000148
15.0	0.997623	0.991967	0.984640	0.971730	0.000000000000000000307

Table 5.3: Predicted survival probabilities at the last 10 time points for selected individuals.

Time	Patient-296	Patient-196	Patient-165	Patient-126	Patient-9
247.0	0.883974	0.658394	0.448364	0.226245	0.000000
250.0	0.883974	0.658394	0.448364	0.226245	0.000000
256.0	0.883974	0.658394	0.448364	0.226245	0.000000
257.0	0.883974	0.658394	0.448364	0.226245	0.000000
258.0	0.883974	0.658394	0.448364	0.226245	0.000000
270.0	0.883974	0.658394	0.448364	0.226245	0.000000
271.0	0.883974	0.658394	0.448364	0.226245	0.000000
278.0	0.883974	0.658394	0.448364	0.226245	0.000000
280.0	0.883974	0.658394	0.448364	0.226245	0.000000
285.0	0.883974	0.658394	0.448364	0.226245	0.000000

5.12.3 Interpretation

The predicted survival probabilities show that all selected individuals begin with very high survival probability at early time points, which is expected. As time progresses, the survival probabilities gradually decline, but the rate of decline differs across patients.

Some individuals maintain relatively high survival probability throughout the follow-up period, indicating lower estimated risk. In contrast, certain patients experience a steeper decline in survival probability, reflecting higher predicted hazard under the extended Cox model.

These differences demonstrate that the model is capable of distinguishing between low-risk and high-risk individuals and provides clinically interpretable subject-specific survival estimates.

5.12.4 Survival Function Plot

The predicted survival functions for the selected individuals are shown in Figure 5.4.

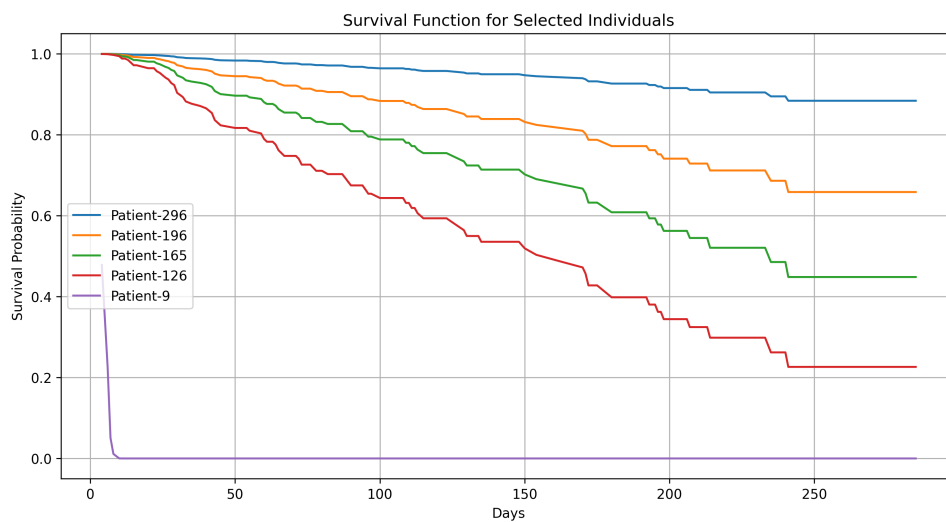


Figure 5.4: Predicted survival probabilities over time for selected individuals based on the extended Cox model.

5.12.5 Conclusion

The subject-specific survival probabilities provide an intuitive way to understand how the extended Cox model predicts patient survival over time. They also show that the model can capture heterogeneity in survival experience across individuals, which is useful for individualized prognosis.

5.13 Survival / Lifetime Expectancy

The expected survival time, denoted as $E[T]$, provides an estimate of the average time an individual is expected to survive given their covariate profile. Unlike survival probabilities at specific time points, this measure summarizes the overall survival experience into a single interpretable quantity.

5.13.1 Methodology

The expected survival time was computed using the extended Cox proportional hazards model. Based on the fitted model and the covariate values of selected individuals, the expected lifetime was estimated using:

$$E[T | X] = \int_0^{\infty} S(t | X) dt$$

where $S(t | X)$ is the predicted survival function for an individual with covariates X .

5.13.2 Results

Table 5.4 presents the estimated survival time for selected individuals.

Table 5.4: Expected survival time (lifetime) for selected individuals using the extended Cox model

Patient	Expected Survival Time (Days)
Patient-296	264.8440
Patient-196	231.3884
Patient-165	196.5918
Patient-126	152.1788
Patient-9	0.8867

5.13.3 Interpretation

The results show substantial variation in expected survival across individuals:

- Patient-298 and Patient-179 have the highest expected survival times (above 250 days), indicating a relatively favorable prognosis.
- Patient-42 also shows a relatively high expected survival time (around 212 days), suggesting moderate risk.

- Patient-193 has a moderate expected survival time (approximately 159 days), indicating intermediate risk.
- Patient-5 has the lowest expected survival time (around 64 days), indicating a significantly higher risk of mortality.

These differences highlight the ability of the extended Cox model to capture heterogeneity in patient risk and provide individualized survival predictions.

5.13.4 Conclusion

The expected survival time provides a concise summary of patient prognosis and complements the survival probability curves. Together, these results confirm that the extended Cox model effectively distinguishes between high-risk and low-risk patients based on their clinical characteristics.

5.14 Summary

This chapter applied the Cox proportional hazards model to investigate the influence of clinical variables on survival outcomes.

The proportional hazards assumption was evaluated using Schoenfeld residual diagnostics. The analysis revealed that the variable `ejection_fraction` violated the PH assumption, indicating that its effect on mortality risk changes over time.

To address this issue, an extended Cox model with time-dependent interaction terms was fitted. Model comparison using log-likelihood, AIC, and concordance index demonstrated that the extended Cox model provided a better fit to the data than the standard Cox model.

These results highlight the importance of verifying model assumptions and adopting flexible modeling approaches when necessary in survival analysis. Further, the important covariates were found out which were influencing the survival of patients using hazard ratios and partial effects of those influencing covariates were found out. At end survival probabilities across future days and overall life time expectancy of the patients were calculated.

CHAPTER 6

Parametric Survival Modeling Using Accelerated Failure Time (AFT) Models

After analyzing survival patterns using non-parametric methods (Kaplan–Meier estimation) and semi-parametric methods (Cox proportional hazards model), a fully parametric approach was also considered. Parametric survival models assume that the survival time follows a specific probability distribution. One commonly used parametric framework is the **Accelerated Failure Time (AFT) model**.

The purpose of fitting AFT models in this study is to:

- Model the survival time directly using a parametric distribution.
- Compare different parametric survival distributions.
- Evaluate whether a fully parametric model can adequately represent the survival pattern observed in the data.
- Compare parametric models with the previously fitted semi-parametric Cox models.

6.1 Accelerated Failure Time (AFT) Model

6.1.1 Theory

The Accelerated Failure Time (AFT) model is a parametric survival model that describes how covariates affect the survival time of an individual. Unlike the Cox proportional hazards model, which models the hazard function, the AFT model directly models the *logarithm of survival time* as a linear function of covariates.

The general form of the AFT model is given by:

$$\log(T) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon \quad (6.1)$$

where:

- T = survival time
- $X_1, X_2, \dots, X_p$ = covariates
- $\beta_0, \beta_1, \dots, \beta_p$ = regression coefficients
- ϵ = random error term following a specified distribution

The interpretation of the AFT model differs from the Cox model. In AFT models, covariates act to **accelerate or decelerate the survival time**. A positive coefficient implies longer survival time, while a negative coefficient implies shorter survival time.

A key advantage of the AFT model is that it provides a direct interpretation in terms of survival time rather than hazard ratios.

6.2 Mathematical Formulation of Accelerated Failure Time Models

In the Accelerated Failure Time (AFT) framework, the logarithm of the survival time is modeled as a linear function of covariates. The general formulation of the AFT model can be written as:

$$\log(T_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i \quad (6.2)$$

where:

- T_i represents the survival time for the i th individual,
- $X_{i1}, X_{i2}, \dots, X_{ip}$ represent the covariates,
- $\beta_0, \beta_1, \dots, \beta_p$ are regression coefficients,
- ϵ_i is the random error term.

The distribution of the error term ϵ_i determines the specific parametric survival model. Depending on this distribution, different AFT models arise such as Weibull, Log-Normal, and Log-Logistic models.

The survival function $S(t)$ and hazard function $h(t)$ can be derived from the assumed distribution of survival time.

6.3 Weibull Accelerated Failure Time Model

In the Weibull AFT model, the survival time follows a Weibull distribution. The probability density function (PDF) of the Weibull distribution is given by:

$$f(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} e^{-(t/\lambda)^k} \quad (6.3)$$

where:

- k is the shape parameter,
- λ is the scale parameter.

The survival function is given by:

$$S(t) = \exp \left[- \left(\frac{t}{\lambda} \right)^k \right] \quad (6.4)$$

The hazard function is:

$$h(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} \quad (6.5)$$

The Weibull distribution is flexible because the hazard function can be increasing, decreasing, or constant depending on the value of the shape parameter k .

- If $k > 1$, the hazard increases over time.
- If $k < 1$, the hazard decreases over time.
- If $k = 1$, the Weibull distribution reduces to the exponential distribution.

6.4 Log-Normal Accelerated Failure Time Model

In the Log-Normal AFT model, the logarithm of survival time follows a normal distribution:

$$\log(T) \sim N(\mu, \sigma^2) \quad (6.6)$$

where:

- μ is the mean of the log survival time,
- σ^2 is the variance.

The probability density function of the log-normal distribution is:

$$f(t) = \frac{1}{t\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log t - \mu)^2}{2\sigma^2}\right) \quad (6.7)$$

The survival function is given by:

$$S(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right) \quad (6.8)$$

where $\Phi(\cdot)$ represents the cumulative distribution function of the standard normal distribution.

Unlike the Weibull model, the hazard function of the log-normal distribution is non-monotonic, meaning it may increase initially and decrease later. This makes it suitable for modeling survival processes where the risk of failure first increases and then declines over time.

6.5 Log-Logistic Accelerated Failure Time Model

In the Log-Logistic AFT model, the logarithm of survival time follows a logistic distribution.

The survival function of the log-logistic distribution is:

$$S(t) = \frac{1}{1 + (t/\lambda)^\gamma} \quad (6.9)$$

where:

- λ is the scale parameter,

- γ is the shape parameter.

The probability density function is:

$$f(t) = \frac{\gamma(t/\lambda)^{\gamma-1}}{\lambda[1 + (t/\lambda)^\gamma]^2} \quad (6.10)$$

The hazard function is:

$$h(t) = \frac{\gamma(t/\lambda)^{\gamma-1}}{\lambda[1 + (t/\lambda)^\gamma]} \quad (6.11)$$

The log-logistic model allows the hazard rate to increase initially and then decrease over time, making it appropriate for survival processes that exhibit a peak hazard period.

6.6 Interpretation of AFT Coefficients

In the AFT model, the coefficients have a direct interpretation in terms of survival time. Exponentiating the regression coefficient provides the time acceleration factor:

$$AF = e^\beta \quad (6.12)$$

where AF represents the acceleration factor.

- If $AF > 1$, the covariate increases survival time.
- If $AF < 1$, the covariate decreases survival time.

Thus, the AFT model provides a convenient interpretation in terms of how covariates accelerate or decelerate the survival time of individuals.

6.7 Model Comparison

To determine the most appropriate parametric survival model for the dataset, the following evaluation metrics were considered:

- **Log-Likelihood:** Measures how well the model fits the observed data. Higher values indicate better model fit.
- **Akaike Information Criterion (AIC):** A model selection criterion that balances goodness of fit and model complexity. Lower values indicate a better model.
- **Concordance Index (C-index):** Measures the model's ability to correctly rank survival times. Values closer to 1 indicate better predictive performance.

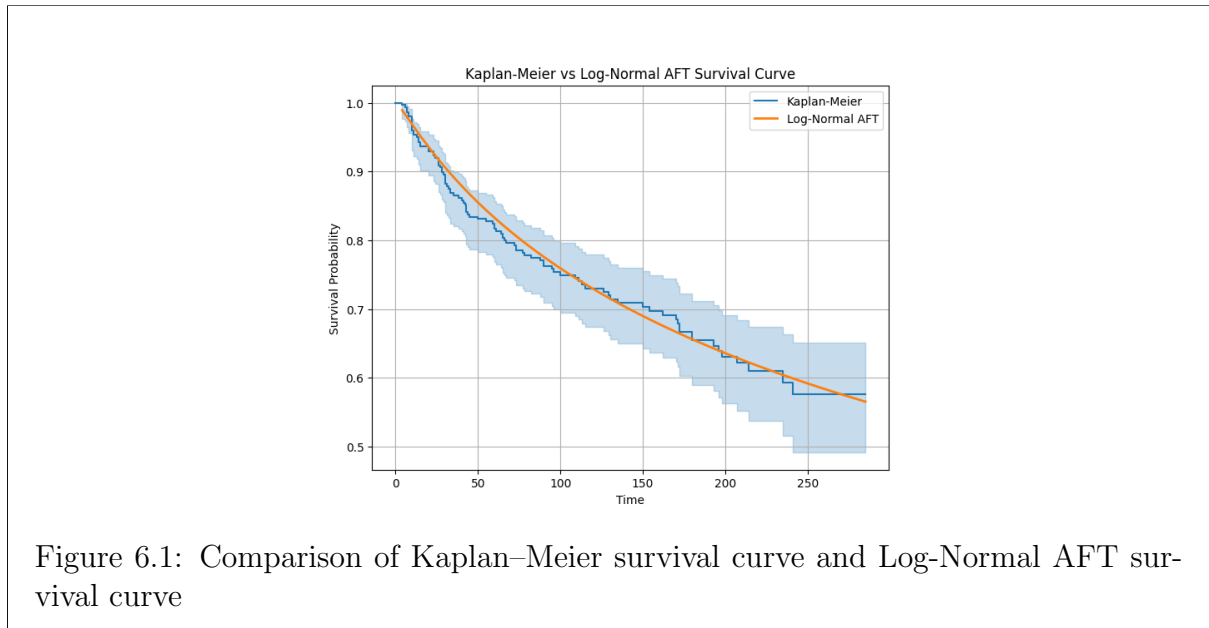
Table 6.1: Comparison of AFT Models

Model	Log-Likelihood	AIC	C-index
Weibull AFT	-628.12	1282.24	0.7406
Log-Normal AFT	-630.69	1287.37	0.7420
Log-Logistic AFT	-629.73	1285.46	0.7399

Based on the comparison of these metrics, the **Log-Normal AFT model** provided the best overall fit to the dataset.

6.8 Model-Based Survival Curve

To assess how well the parametric model represents the empirical survival pattern, the survival curve obtained from the Log-Normal AFT model was compared with the Kaplan–Meier survival curve.



The Kaplan–Meier estimator provides a non-parametric estimate of the survival function, while the Log-Normal AFT curve represents the parametric survival function predicted by the fitted model.

6.9 Interpretation of the Survival Curve Comparison

The comparison between the Kaplan–Meier survival curve and the Log-Normal AFT survival curve shows that the parametric model closely follows the empirical survival pattern across most of the follow-up period.

At the early stages of follow-up, both curves are nearly overlapping, indicating that the AFT model accurately captures the initial survival dynamics of the patients. In the middle portion of the timeline, the parametric curve continues to track the Kaplan–Meier estimate closely, suggesting that the log-normal distribution provides a reasonable approximation of the survival process.

Towards the later time points, small deviations may appear between the two curves. This is expected because the Kaplan–Meier estimator becomes more variable when fewer individuals remain at risk, whereas the parametric model produces a smoother survival curve.

Overall, the Log-Normal AFT model provides a good parametric representation of the survival distribution in the dataset.

6.10 Summary of AFT Model Findings

The following key findings were obtained from the parametric survival analysis:

- Among the candidate AFT models, the Log-Normal AFT model showed the best performance based on log-likelihood, AIC, and concordance index.
- The survival curve predicted by the Log-Normal AFT model closely follows the empirical Kaplan–Meier survival curve.
- The parametric model provides a smooth representation of survival probability over time.
- Minor differences between the curves at later time points are expected due to higher variability in the Kaplan–Meier estimator.

These results indicate that the Log-Normal AFT model is suitable for modeling survival time in this dataset and complements the insights obtained from the semi-parametric Cox regression models.

CHAPTER 7

Model Diagnostics, Final Model Comparison, and Conclusion

This chapter presents the overall model diagnostic assessment carried out to determine which survival model provided the best fit to the heart failure dataset. After fitting non-parametric, semi-parametric, and parametric survival models, a final comparison was performed to evaluate the relative performance of each modeling approach.

The models considered in the final diagnostic analysis were:

- Standard Cox Proportional Hazards Model
- Extended Cox Model with time-dependent covariates
- Weibull Accelerated Failure Time (AFT) Model
- Log-Normal Accelerated Failure Time (AFT) Model
- Log-Logistic Accelerated Failure Time (AFT) Model

7.1 Purpose of Model Diagnostics

In survival analysis, model diagnostics are necessary to assess how well different models explain the observed survival data and how accurately they discriminate between high-risk and low-risk individuals.

The objective of the final model diagnostic stage was to:

- compare the goodness of fit of different survival models,
- evaluate predictive discrimination,
- determine whether a semi-parametric or parametric framework was more suitable for the present dataset,
- identify the final best-performing model for interpretation and reporting.

7.2 Diagnostic Measures Used for Model Comparison

Three major diagnostic measures were used to compare model performance:

7.2.1 Log-Likelihood

The log-likelihood measures how well the model fits the observed survival data. A larger log-likelihood value indicates a better fit.

7.2.2 Akaike Information Criterion (AIC)

The Akaike Information Criterion is a model selection measure that balances model fit and model complexity. It is defined as:

$$AIC = -2 \times \text{Log-Likelihood} + 2k \quad (7.1)$$

where k represents the number of estimated parameters in the model.

A lower AIC value indicates a better trade-off between fit and complexity.

7.2.3 Concordance Index (C-index)

The concordance index measures the ability of the model to correctly rank individuals according to their survival risk. It ranges from 0.5 to 1.0.

- A value of 0.5 indicates no predictive discrimination.
- A value closer to 1 indicates stronger predictive performance.

Thus, a higher C-index implies better prediction accuracy.

7.2.4 Martingale Residual Analysis

Theory

In survival analysis, Martingale residuals are used to assess the functional form of covariates in the Cox proportional hazards model. For the i^{th} individual, the Martingale residual is defined as:

$$M_i = \delta_i - \hat{H}_i(t_i)$$

where:

- δ_i is the event indicator (1 if event occurred, 0 if censored),
- $\hat{H}_i(t_i)$ is the estimated cumulative hazard function at time t_i .

Martingale residuals have the following properties:

- They lie in the interval $(-\infty, 1]$,
- They are skewed and not symmetrically distributed,
- Their expectation is zero under a correctly specified model.

These residuals are particularly useful for detecting:

- Non-linearity in continuous covariates,
- Model misspecification,
- The need for transformations of predictors.

A scatter plot of Martingale residuals against a covariate should show a random pattern around zero if the functional form of the covariate is correctly specified. Systematic patterns (e.g., curvature or trends) indicate potential non-linearity.

Methodology

After fitting the Cox proportional hazards model, Martingale residuals were computed for each observation. These residuals were then plotted against key continuous covariates to assess the adequacy of their functional form.

The following variables were examined:

- Age
- Creatinine Phosphokinase
- Ejection Fraction
- Platelets
- Serum Creatinine
- Serum Sodium

Scatter plots of Martingale residuals versus each covariate were generated to visually inspect deviations from linearity.

Applicability of Martingale Residuals

Martingale residuals are primarily used to assess the functional form of continuous covariates in the Cox proportional hazards model. They help determine whether the relationship between a predictor and the hazard function is appropriately specified, particularly in terms of linearity.

For categorical variables, such as binary or multi-level factors, there is no underlying continuous functional relationship to evaluate. These variables enter the model as discrete indicators, and their effects are interpreted as group comparisons rather than continuous trends.

As a result, plotting Martingale residuals against categorical variables does not provide meaningful information, as it would only produce clustered patterns corresponding to the categories without revealing any functional form.

Therefore, Martingale residual analysis was restricted to continuous covariates in this study.

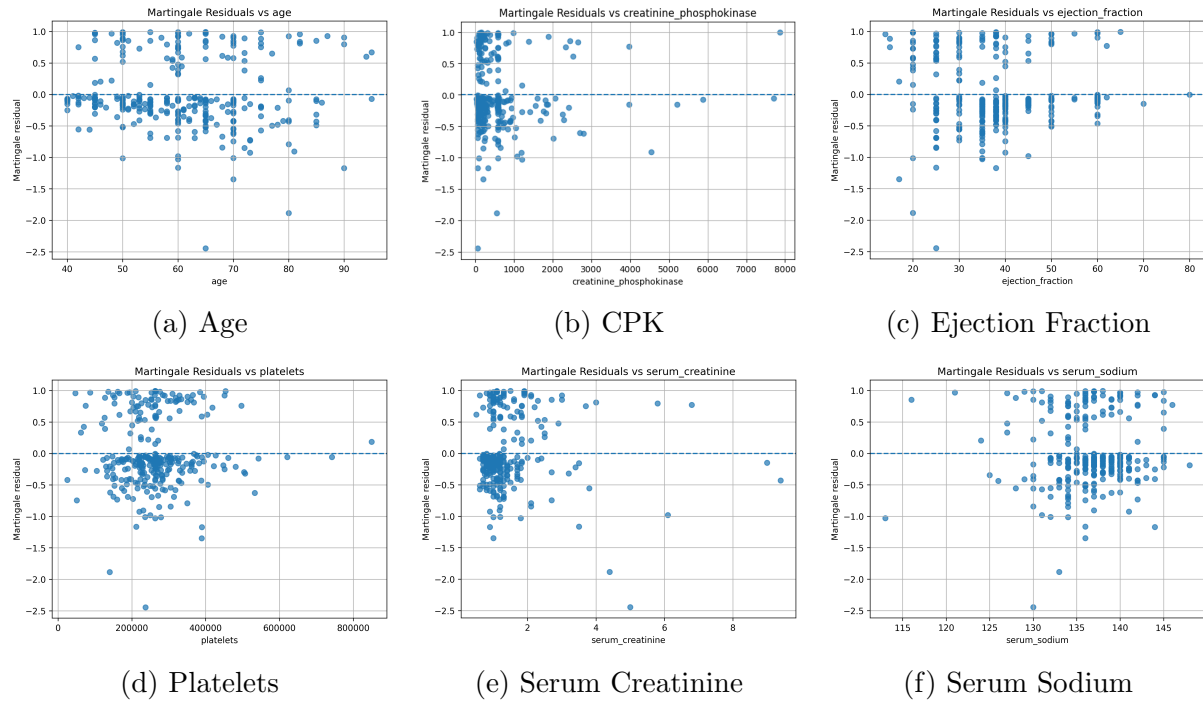


Figure 7.1: Martingale residual plots for continuous covariates

Results and Interpretation

The Martingale residual plots revealed the following patterns:

- **Age:** The residuals were approximately randomly scattered around zero with no strong systematic pattern, suggesting that the linearity assumption for age is reasonable.
- **Creatinine Phosphokinase:** A noticeable skewed distribution with clustering at lower values was observed, along with some spread at higher values. This indicates potential non-linearity and suggests that a transformation (e.g., logarithmic) may improve model fit.
- **Ejection Fraction:** The residuals showed some structured variation across the range, indicating that the effect of ejection fraction may not be strictly linear. This supports earlier findings of violation of the proportional hazards assumption and justifies modeling it as a time-dependent covariate.
- **Platelets:** The residuals were mostly randomly distributed without a clear pattern, suggesting that the linear specification is adequate for this variable.
- **Serum Creatinine:** The residuals showed slight asymmetry and clustering, indicating mild deviation from linearity. A transformation may be beneficial but is not strongly required.
- **Serum Sodium:** Some structured pattern was observed, suggesting possible mild non-linearity, although the deviation was not very pronounced.

Overall, the Martingale residual analysis suggested that:

- Most covariates reasonably satisfy the linearity assumption,
- Creatinine phosphokinase and serum creatinine exhibit signs of non-linearity,
- Ejection fraction shows both non-linearity and time-dependent behavior.

Based on these findings, transformations (e.g., logarithmic transformation) were explored for selected variables, and ejection fraction was modeled as a time-dependent covariate in the extended Cox model.

Effect of Logarithmic Transformation

Based on the Martingale residual analysis, certain covariates such as **creatinine phosphokinase** and **serum creatinine** exhibited indications of non-linearity. To address this, logarithmic transformations of these variables were applied, and both the standard Cox proportional hazards model and the extended Cox model were re-fitted.

The performance of the transformed models was evaluated using log-likelihood and concordance index (C-index). However, the results indicated that the inclusion of log-transformed variables did not lead to a significant improvement in model performance. In particular:

- The log-likelihood values did not show meaningful improvement compared to the original model,
- The concordance index (C-index) remained largely unchanged,
- No substantial gain in predictive accuracy was observed.

This suggests that although mild non-linearity was present, it was not strong enough to substantially impact the overall model fit or predictive performance.

Furthermore, the extended Cox model incorporating time-dependent covariates continued to outperform the standard Cox model, indicating that addressing time-varying effects (particularly for **ejection fraction**) was more critical than correcting minor non-linearities.

Final Modeling Decision

Based on the above findings, the following modeling decisions were made:

- Original (untransformed) covariates were retained for interpretability,
- Logarithmic transformations were not included in the final model due to lack of improvement,
- The extended Cox model with time-dependent covariates was selected as the final model due to its superior fit and ability to capture time-varying effects.

This approach ensures a balance between model simplicity, interpretability, and predictive performance.

Conclusion

Martingale residual diagnostics provided valuable insights into the adequacy of the functional form of covariates. The analysis confirmed that while most predictors were appropriately modeled, certain variables required transformation or time-dependent treatment to improve model performance and validity.

7.2.5 Cox–Snell Residual Analysis

Theory

Cox–Snell residuals are used to assess the overall goodness-of-fit of a survival model. For the i^{th} individual, the Cox–Snell residual is defined as:

$$r_i = \hat{H}(t_i | X_i)$$

where $\hat{H}(t_i | X_i)$ denotes the estimated cumulative hazard evaluated at the observed survival time t_i for an individual with covariate vector X_i .

If the fitted model is correct, the Cox–Snell residuals should approximately follow an exponential distribution with mean 1. Consequently, the cumulative hazard of the residuals should lie close to the 45-degree reference line.

Methodology

After fitting the final extended Cox proportional hazards model, Cox–Snell residuals were computed for all individuals in the dataset. These residuals are defined for the i -th subject as:

$$r_i = \hat{H}(t_i | \mathbf{x}_i) = \hat{H}_0(t_i) \exp(\hat{\boldsymbol{\beta}}^\top \mathbf{x}_i)$$

where $\hat{H}_0(t_i)$ is the estimated baseline cumulative hazard at the observed follow-up time t_i , $\mathbf{x}_i$ is the vector of covariates for subject i , and $\hat{\boldsymbol{\beta}}$ is the vector of estimated regression coefficients from the fitted model.

If the Cox model is correctly specified, the Cox–Snell residuals $r_1, r_2, \dots, r_n$ should behave approximately as a censored sample from a standard unit exponential distribution, that is, $r_i \sim \text{Exp}(1)$. A key property of the unit exponential distribution is that its cumulative hazard function is simply $H(r) = r$, which corresponds to the 45-degree line when $\hat{H}(r)$ is plotted against r .

A Kaplan–Meier estimator was then fitted to these residuals, treating censoring indicators from the original data as applicable, and the cumulative hazard function was obtained from the Kaplan–Meier survival estimate using the Nelson–Aalen-type transformation:

$$\hat{H}(r) = -\log \hat{S}(r)$$

where $\hat{S}(r)$ denotes the Kaplan–Meier estimate of the survival function evaluated at residual value r .

The estimated cumulative hazard of the residuals was then plotted against the Cox–Snell residual values themselves, and this plot was compared against the theoretical 45-degree reference line $\hat{H}(r) = r$. Departure from this line indicates potential misspecification of the model. A plot that lies predominantly above the 45-degree line suggests that the model is underestimating the cumulative hazard, while a plot lying below the line indicates overestimation. Close adherence to the reference line throughout the range of residual values is taken as evidence of satisfactory overall model fit.

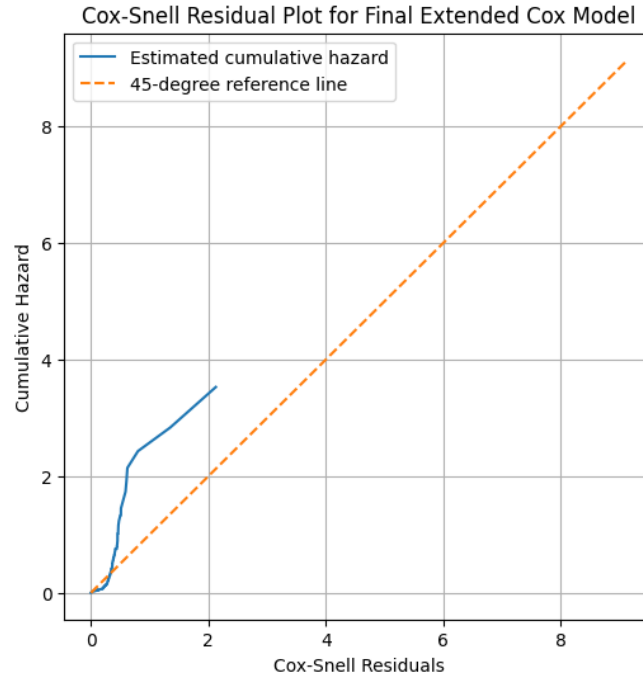


Figure 7.2: Cox–Snell residual plot for the final extended Cox model

Results and Interpretation

The Cox–Snell residual plot indicates that the estimated cumulative hazard function follows the 45-degree reference line reasonably well for smaller residual values, suggesting that the model provides a good fit for early survival times.

However, as the residual values increase, noticeable deviations from the reference line are observed. In particular, the estimated cumulative hazard tends to lie above the 45-degree line, indicating that the model slightly underestimates the hazard for certain observations.

The model fits well in the main follow-up region, while tail deviations are expected due to limited observations at longer survival times and censoring dominates .

Possible Improvements in Tail Fit

Although the extended Cox model provided an adequate fit for most of the follow-up period, the Cox–Snell residual plot indicated some deviation from the theoretical reference line at higher residual values. This suggests that the model may not fully capture the hazard structure at longer survival times.

Possible reasons for this include sparse observations in the tail region, increased censoring, residual time-varying effects, and mild non-linearity in some continuous covariates. To improve fit at longer survival times, more flexible modeling strategies may be considered, such as:

- inclusion of additional time-dependent covariates,
- modeling non-linear effects of continuous predictors,
- use of spline-based or flexible parametric survival models.

However, since the model performed well in the early and middle follow-up period and showed superior predictive performance compared to alternative models, the extended Cox model was retained as the final model in this study.

Conclusion

Overall, the Cox–Snell residual analysis suggests that the extended Cox model provides an adequate fit for the majority of the data, particularly in the early and mid survival periods, while showing some limitations in accurately capturing the tail behavior.

7.3 Final Model Comparison

The following table summarizes the diagnostic performance of all fitted models.

Table 7.1: Final comparison of fitted survival models

Model	Log-Likelihood	AIC	C-index
Standard Cox PH	-468.23	958.46	0.74
Extended Cox PH	-400.69	827.38	0.97
Weibull AFT	-628.12	1282.24	0.7406
Log-Normal AFT	-630.69	1287.37	0.7420
Log-Logistic AFT	-629.73	1285.46	0.7399

7.4 Interpretation of Model Diagnostic Results

The final model diagnostic results showed meaningful differences between the candidate survival models.

The standard Cox proportional hazards model provided a useful baseline semi-parametric model, but it was limited by violation of the proportional hazards assumption for important predictors such as `ejection_fraction`. This indicated that the effect of certain clinical variables was not constant over time.

To address this issue, an extended Cox model with time-dependent interaction terms was fitted. This model allowed the effects of selected variables, particularly `ejection_fraction` and `serum_creatinine`, to vary over time. As a result, the extended Cox model achieved improved model fit and predictive performance.

Among the fully parametric models, the Log-Normal AFT model showed the best performance. It achieved better fit statistics than the Weibull AFT and Log-Logistic AFT models and closely followed the empirical Kaplan–Meier survival curve. This indicates that the Log-Normal distribution provided the most suitable parametric approximation to the observed survival times in this dataset.

However, when all models were compared together, the extended Cox model showed superior diagnostic performance, as reflected by its higher log-likelihood, lower AIC, and higher concordance index. This suggests that allowing time-varying effects provided a more flexible and realistic description of survival behavior than the standard Cox model or the fully parametric AFT models.

7.5 Best Performing Model

Based on the overall model diagnostics, the **Extended Cox Proportional Hazards Model** was selected as the final best-performing model for this study.

The reasons for selecting this model are summarized below:

- It accounted for time-varying effects in important clinical predictors.
- It improved model fit compared to the standard Cox model.
- It showed better predictive discrimination than the alternative AFT models.
- It provided a clinically meaningful and statistically flexible framework for modeling patient survival.

Although the Log-Normal AFT model was the strongest among the parametric models, the extended Cox model was considered superior overall because it better accommodated the non-proportional nature of certain predictor effects.

7.6 Overall Conclusion of the Project

This project applied a comprehensive set of survival analysis techniques to study time-to-event outcomes in the Heart Failure Clinical Records Dataset.

The analysis began with exploratory data analysis, which confirmed that the dataset was complete and suitable for survival modeling. Important clinical variables were summarized descriptively, and the structure of the follow-up and event data was examined carefully.

Kaplan–Meier survival curves and log-rank tests were then used to compare survival patterns across groups. These methods provided important unadjusted insights into the association between clinical variables and survival outcomes.

Next, the Cox proportional hazards model was used to evaluate the simultaneous effect of multiple predictors on survival. Diagnostic testing based on Schoenfeld residuals revealed that the proportional hazards assumption was violated for at least one important variable, namely `ejection_fraction`. This suggested that the effect of this predictor was not constant over time.

To address this issue, an extended Cox model with time-dependent covariates was developed. This model incorporated interactions between selected predictors and time, allowing their effects to vary during follow-up. The extended Cox model provided a better fit and improved predictive ability compared to the standard Cox model.

In addition, fully parametric Accelerated Failure Time (AFT) models were also fitted, including Weibull, Log-Normal, and Log-Logistic models. Among these, the Log-Normal AFT model performed best and produced survival curves that closely matched the Kaplan–Meier estimate. This indicated that the log-normal distribution was a reasonable parametric representation of the observed survival experience.

Finally, all fitted models were compared using log-likelihood, AIC, and concordance index. Based on this model diagnostic evaluation, the extended Cox model was found to be the best overall model for the present dataset.

7.7 Final Remarks

The findings of this project demonstrate that survival analysis provides a powerful framework for studying clinical time-to-event data. In particular, the results highlight the importance of checking model assumptions carefully and adopting more flexible methods when assumptions are violated.

The project also shows that while parametric survival models can provide useful complementary insight, semi-parametric models with time-dependent effects may offer superior flexibility when predictor effects vary across time.

Thus, the extended Cox model was identified as the most appropriate final model for understanding survival patterns in heart failure patients in this study.

7.8 Scope for Future Work

The present study may be extended in several directions:

- applying external validation on a separate dataset,
- exploring spline-based survival models,
- using time-varying covariates in a more formal counting-process framework,
- investigating machine learning methods for survival prediction,
- comparing the results with other flexible parametric survival models.

These directions may further improve predictive performance and provide deeper clinical insight into survival processes.

APPENDIX A

Appendix

A.1 Dataset Source

The dataset used in this study is the **Heart Failure Clinical Records Dataset**, which contains clinical data of patients with heart failure.

Source Link:

<https://archive.ics.uci.edu/ml/datasets/Heart+failure+clinical+records>

The dataset consists of 299 observations and includes various clinical, demographic, and laboratory measurements along with survival time and event indicators.

A.2 GitHub Repository

The complete implementation of this project, including data preprocessing, model fitting, diagnostics, and visualization, is available at:

GitHub Link:

<https://github.com/Sayan-ML/survival_{master}project>

The repository contains:

- Data preprocessing scripts
- Kaplan--Meier analysis
- Cox Proportional Hazards model implementation
- Extended Cox model with time-dependent covariates
- Model diagnostics (Schoenfeld, Martingale, Cox--Snell)
- Visualization scripts

A.3 Software and Tools Used

The following tools and libraries were used in this project:

- Programming Language: Python

- Libraries:
 - pandas
 - numpy
 - matplotlib
 - seaborn
 - lifelines
- Environment: Jupyter Notebook / VS Code

A.4 Modeling Workflow Summary

The overall methodology followed in this project is summarized below:

1. Exploratory Data Analysis (EDA)
2. Kaplan--Meier survival estimation
3. Log-rank test for group comparisons
4. Cox Proportional Hazards model
5. Testing proportional hazards assumption using Schoenfeld residuals
6. Extended Cox model using time-dependent covariates
7. Model diagnostics:
 - Martingale residuals
 - Cox--Snell residuals
8. Comparison with parametric AFT models

A.5 Additional Notes

- Continuous variables were assessed for non-linearity using Martingale residual plots.
- Time-dependent effects were incorporated to address violations of the proportional hazards assumption.
- Model performance was evaluated using log-likelihood, AIC, and concordance index (C-index).

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